

A Micro Anatomy of the Hidden Local Government Debt in China: 2014-2023*

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Abstract

This paper introduces a newly constructed dataset comprising firm-level and city-level information on 2,940 industrial Local Government Financing Vehicles (LGFVs) across 305 cities in China. Compiled from public disclosures, the dataset includes detailed financial metrics such as assets, liabilities, profitability, and ownership structure at the firm level, alongside key economic indicators at the city level. We document significant heterogeneity in hidden debt exposures across regions and sectors. The dataset supports applications in studying geographic patterns of off-balance-sheet debt and firm-level financial behaviors, providing valuable insights for future research into fiscal transparency and debt sustainability in China.

Keywords: Local Government Debt

JEL Classification Numbers: H74, G32, E62

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1 Introduction

Local government debt in China has experienced unprecedented expansion, with officially reported on-balance-sheet debt increasing dramatically from 7.13% to 32.18% of GDP during the 2015–2023 period.¹ This growth trajectory stands in stark contrast to international patterns, exemplified by the United States, where local government debt peaked at 11.51% of GDP during the 2008–2009 financial crisis before declining to 7.2% by 2023.²

The rapid debt expansion in China primarily reflects the systematic migration of previously opaque off-balance-sheet borrowing onto official government accounts, driven by increasingly stringent central government regulations implemented after 2015. However, the existing academic literature predominantly relies on aggregated provincial-level data, employs outdated analytical frameworks, or focuses exclusively on on-budget liabilities. Consequently, the magnitude, structural composition, and temporal evolution of off-budget borrowing through Local Government Financing Vehicles (LGFVs) remain insufficiently understood, particularly during the critical post-2015 fiscal reform period.

Several fundamental empirical questions demand rigorous investigation: Following the implementation of the revised Budget Law in 2015, what constitutes the actual magnitude of total local government debt, including previously hidden off-balance-sheet liabilities? How substantial is the hidden debt component relative to official figures, and how does this debt distribute across geographic regions and administrative levels? Understanding heterogeneity at both city and firm levels represents an essential prerequisite for conducting rigorous macroeconomic research on fiscal policy effectiveness and systemic financial risks.

This paper makes three primary contributions to the literature. First, we construct a comprehensive, research-ready dataset that systematically integrates detailed financial statements

¹Provincial-level data on China’s on-balance-sheet local government debt are sourced from the ([China Electronic Local Government Bond Market Access platform](#)).

²U.S. local government debt statistics are obtained from the [Budget of the United States Government](#). A comprehensive comparison of government debt levels and growth trajectories between China and the United States (2015–2023) is presented in Appendix [A.1](#).

from Chinese LGFVs with extensive city-level fiscal and macroeconomic indicators. Spanning the period from 2014 to 2023, this dataset encompasses 2,940 LGFVs across 305 cities, providing unprecedented geographical and temporal coverage alongside granular financial metrics. Second, we establish clear methodological distinctions between officially reported on-budget debt and off-budget liabilities raised through LGFVs, offering detailed breakdowns by debt structure, financing channels, and regional characteristics. Third, our analysis reveals significant stylized facts regarding the evolution of China’s local government financing landscape.

The dataset establishes an important empirical foundation for addressing diverse research questions, including the determinants of hidden local government debt accumulation, relationships between debt composition and long-term fiscal sustainability, and complex interactions between local debt dynamics and real estate market fluctuations. While this paper emphasizes data construction and descriptive analysis rather than causal identification, our findings document substantial expansion of hidden local government debt—from ¥10.2 trillion (\$1.5 trillion) in 2014 to ¥50.9 trillion (\$8.6 trillion) in 2023. This growth results in total local government debt-to-GDP ratios (combining on-balance-sheet and off-balance-sheet components) surging from 15.61% to 66.93% over the study period. Furthermore, our analysis uncovers notable geographic heterogeneity in debt structures and financing strategies, with inland cities demonstrating greater reliance on hidden borrowing mechanisms and strategic balancing between formal and informal financing channels.

This research advances the analytical framework from macro-level aggregates to micro-level financial data examination. By constructing a novel city-level dataset, we address a critical data gap in the existing literature while offering new insights into local government fiscal behavior and systemic risks associated with China’s infrastructure-driven growth model. These findings contribute meaningfully to ongoing policy discussions regarding fiscal transparency, local government incentive structures, and debt sustainability challenges in emerging market contexts.

The remainder of this paper proceeds as follows. Section 2 provides comprehensive back-

ground and institutional context regarding local government debt mechanisms in China. Section 3 describes our dataset construction methodology and presents detailed discussion of off-balance-sheet debt measurement. Section 4 explains our data validation procedures through decomposition of national-level temporal trends. Section 5 presents city-level analysis of local government debt patterns, emphasizing geographical distribution and structural transformations over time. Section 6 extends the investigation to firm-level analysis, examining financial performance metrics and leverage trends across different enterprise categories.

2 Background

The rapid expansion of China’s local government debt, particularly its hidden components, traces its origins to the 2008–2009 global financial crisis response. Confronting a sharp decline in export demand, China implemented a massive fiscal stimulus package totaling approximately ¥4 trillion (roughly \$586 billion USD). The central government directly provided only 30% of this funding, leaving local governments responsible for financing the remaining 70% (?). However, China’s fiscal institutional framework, established following the 1994 tax-sharing reforms (fenshuizhi), imposed strict borrowing constraints on subnational authorities (?).³

Before 2015, local governments faced legal prohibition from direct borrowing or operating budget deficits, despite bearing substantial expenditure responsibilities for infrastructure development, public services, and economic development initiatives (?). This institutional arrangement created a fundamental mismatch between fiscal capacity and expenditure obligations, necessitating innovative financing mechanisms.

To circumvent these regulatory restrictions, local governments increasingly established and utilized Local Government Financing Vehicles (LGFVs)—special-purpose corporate entities de-

³The 1994 tax-sharing reform (fenshuizhi) fundamentally restructured China’s fiscal system by centralizing tax collection while maintaining expenditure responsibilities at local levels, creating the structural fiscal imbalance that necessitated alternative financing mechanisms.

signed to facilitate off-budget borrowing.⁴ LGFVs raised funds through multiple channels, including bank loans, corporate bond issuances, trust loans, and wealth management products, typically secured by land-use rights or backed by implicit government guarantees (?). This indirect borrowing mechanism—commonly referred to as "hidden debt" (or off-sheet debt)—significantly expanded local fiscal capacity outside formal budget constraints while simultaneously creating substantial risks to financial stability and long-term fiscal sustainability (?).

Land finance has significantly contributed to the rise of off-sheet local government debt in China. By systematically transferring land-use rights to Local Government Financing Vehicles (LGFVs), local governments leverage land assets to secure financing for infrastructure projects, enhancing land values and stimulating economic development. This practice has created a strong interdependence between local fiscal stability and volatile real estate markets, fueling a high-leverage financing cycle (???).⁵

The proliferation of hidden local government debt has attracted substantial attention within academic and policy communities. Empirical studies emphasize that extensive local government borrowing can crowd out private investment opportunities, thereby reducing overall economic efficiency and undermining long-term growth potential (?). Additionally, widespread implicit government guarantees distort market-based risk pricing mechanisms and amplify systemic financial risks, raising legitimate concerns about potential financial instability (?). Recent comprehensive audits and international assessments estimate China's hidden LGFV debt stock at approximately ¥60 trillion by 2023, equivalent to nearly 50% of GDP.⁶ Policymakers have responded through debt restructuring programs and institutional reforms aimed at improving fiscal transparency and discipline, yet fundamental fiscal imbalances between revenues and expenditure responsibil-

⁴Local Government Financing Vehicles are off-budget entities wholly or partially owned by local governments that finance infrastructure and public services through bank loans, bonds, and other instruments, often backed by land or implicit guarantees.

⁵Land finance involves monetizing public land assets through sales or leases to generate revenue, facilitating debt accumulation via LGFVs. Post-2010 regulatory constraints led LGFVs to increasingly rely on shadow banking (such as trust loans and wealth management products).

⁶International Monetary Fund estimates suggest total LGFV liabilities reached approximately ¥60 trillion (\$8.4 trillion USD) by 2023, though official Chinese government figures report substantially lower amounts, highlighting measurement challenges in this area.

ities remain largely unresolved (?).

Although China’s LGFV-driven debt expansion is unprecedented in scale and institutional complexity, similar off-budget financing mechanisms exist globally, making China’s experience internationally relevant. Local governments in both developing and advanced economies have adopted analogous vehicles for infrastructure financing, typically relying on implicit public guarantees while operating under soft budget constraints (?).⁷ These arrangements typically carry implicit expectations of public bail-outs when difficulties arise, suggesting that China’s experience provides valuable insights into the fiscal and institutional risks of hidden local government liabilities under decentralization and investment pressure.

Building upon this institutional context and existing literature, this paper contributes to ongoing academic and policy debates by presenting a newly constructed, comprehensive dataset covering 2,940 LGFVs operating during the period following China’s revised Budget Law implementation in 2015. This legislation officially permitted local governments to issue municipal bonds for the first time, fundamentally altering the institutional landscape for local government finance. Our dataset systematically links detailed firm-level financial metrics—including comprehensive asset and liability compositions, profitability indicators, and ownership structures—with corresponding city-level fiscal and macroeconomic variables, enabling unprecedented granular analysis of hidden debt phenomena. By documenting substantial heterogeneity across regions and sectors, this dataset enhances scholarly understanding of local fiscal behaviors, political economy incentives underlying off-budget borrowing decisions, and broader implications for credit allocation efficiency. Ultimately, this paper’s primary data contribution supports more transparent and evidence-based policy evaluation processes, while highlighting essential structural fiscal reforms necessary for aligning local government revenue capacities with expenditure respon-

⁷Developing countries examples include India’s special-purpose vehicles (SPVs) and public-sector undertakings (PSUs) backed by state transfers or guarantees (?); Brazil’s state development banks and quasi-public companies funding infrastructure through contingent liabilities (?); and Vietnam’s provincial Local Development Investment Funds authorized to raise off-budget capital under local government oversight (?). Advanced economies similarly employ mechanisms such as special districts and tax-increment financing (TIF) in the United States (??), and public-private partnerships (PPPs) and project-specific SPVs in the United Kingdom (??).

sibilities and reducing systemic financial risks associated with China’s persistent hidden debt challenges.

3 Data

This paper integrates multiple data sources to construct a comprehensive dataset linking bond market activities with firm performance across Chinese cities. Primary data sources include the Wind Financial Terminal (WIND), China Stock Market & Accounting Research (CSMAR), and China’s City Statistical Yearbooks. A detailed list of firm-level and city-level variables is provided in the Appendix [A.2](#).

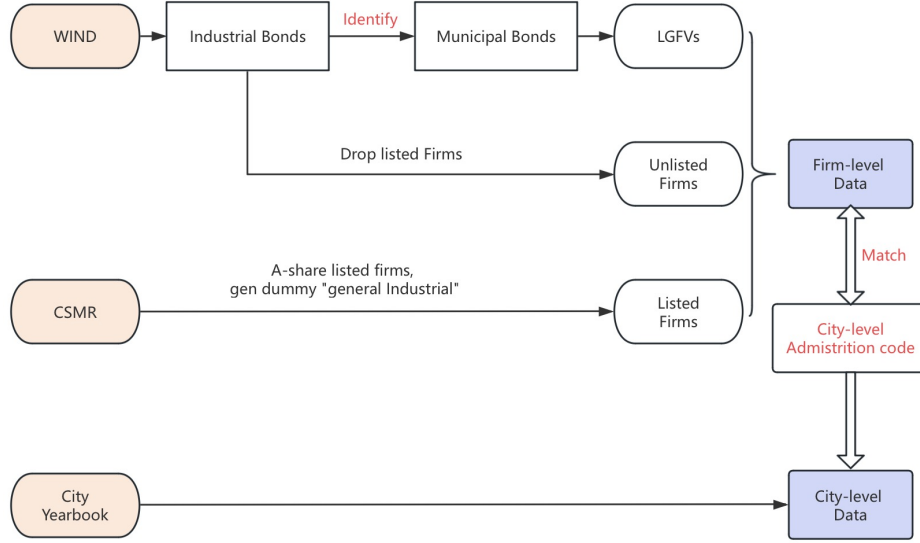
Total local government debt is decomposed into on-sheet and off-sheet local government debts. On-sheet debt data are obtained from China’s City Statistical Yearbooks, while off-sheet debt data are sourced from the WIND database. Specifically, we define off-sheet debt as long-term bank loans and bonds payable issued by Local Government Financing Vehicles (LGFVs) that have secured financing through municipal bond issuances. This conservative measure excludes other potential liabilities of LGFVs not explicitly backed by local governments.

City-level socioeconomic and fiscal indicators, including GDP, population, fiscal revenue, and official debt figures, are sourced from the *China City Statistical Yearbook*, published annually by the National Bureau of Statistics.

3.1 Dataset Construction Process

As illustrated in Figure [1](#), our data construction involves sequential steps to integrate multiple sources into a unified analytical framework. The process begins with industrial bond data extraction from the WIND database. From this comprehensive bond dataset, we identify municipi-

Figure 1: Data Construction



pal bonds according to ChinaBond classification standards.⁸ Companies issuing these municipal bonds are subsequently classified as Local Government Financing Vehicles (LGFVs).

Firm-level data construction follows two parallel tracks. For listed firms, we utilize the CSMAR database, extracting data on A-share listed companies and creating binary indicators to classify industrial sector firms per China Securities Regulatory Commission (CSRC) standards. For unlisted firms, we gather data from industrial bond issuers recorded in the WIND database, explicitly excluding firms already listed in CSMAR to prevent duplication.

In the final construction stage, critical matching procedures consolidate LGFV and unlisted firm data into a unified firm-level dataset. Subsequently, we match these firm-level records to standardized city-level administrative codes, enabling robust geographic analysis of financing behaviors and firm outcomes. Administrative code mappings are sourced from a comprehensive open-source repository,⁹ ensuring consistent geographical identification across time periods.

⁸Alternative definitions of LGFV bonds and municipal bonds are discussed in Appendix A.3.1. [chen2020financing] employs this definition. This classification has been available since 2013, though our analysis covers 2014–2023 to capture more comprehensive LGFV financial information during the post-2015 reform period.

⁹Administrative divisions data are obtained from the public Github project [Administrative-divisions-of-China project](#), providing consistent mappings of administrative units and documenting historical boundary changes.

3.2 Local Government Debt in China

Existing literature offers various estimates of China’s total local government debt, yet comprehensive city-level time series data—especially after 2015—remain unavailable. This paper addresses that gap by constructing such a dataset. Off-sheet debt, largely accumulated through Local Government Financing Vehicles (LGFVs), refers to obligations not formally recorded in government accounts. Current estimation approaches focus on LGFV-incurred liabilities, typically using one of three main methods:

1. **LGFV-level financial statements:** Researchers obtain comprehensive lists and financial statements from LGFVs. [huang2020local], for example, defines off-sheet debt as the sum of bank loans and bonds issued by LGFVs, reporting a growth from RMB 1.2 trillion in 2006 to RMB 12.5 trillion in 2013, with bank loans constituting about 79%. Similarly, [bai2018financial] uses financial reports from approximately 1,800 LGFVs, estimating total liabilities at roughly RMB 46 trillion by 2015.
2. **Funding sources, particularly local banks:** Direct data on LGFV bank lending is scarce due to confidentiality. Nonetheless, indirect measures, such as the National Audit Office’s (NAO) 2013 audit, offer partial insights. [li2024crowding] utilized confidential bank-side data, emphasizing the impact of China’s 2015 debt-to-bond swap policy, which redirected credit towards private enterprises and enhanced local productivity.
3. **Bond market data:** Many studies leverage publicly available bond issuance data to estimate outstanding LGFV bonds. This method is popular due to data accessibility, ease of processing, and widespread acceptance within the literature. Prominent examples include [oi2025perfect], [chen2020financing], and [shih2024local], all relying extensively on the WIND database, the most comprehensive bond data provider in China. However, due to inconsistencies in city-level identifiers in WIND, most analyses aggregate data at provincial or national levels, sacrificing city-level granularity.

This paper aligns closely with the first methodological approach. Specifically, we integrate data on loans and bonds from the Wind database, matching these with detailed city-level statistics from local government statistical yearbooks. Following [huang2020local] and [bai2018financial], we measure off-sheet debt by identifying LGFV-issued bonds, especially focusing on the bonds issued by the LGFVs¹⁰ using the ChinaBond classification. Detailed definitions and further discussion on off-sheet debt, LGFVs, and LGFV bonds are provided in Appendix A.3.

3.3 Dataset Overview and Summary Statistics

Our integrated dataset combines three primary data sources covering 2014–2023: city-level fiscal indicators from statistical yearbooks, corporate bond information from WIND, and listed company data from CSMAR. The final dataset identifies 2,940 LGFVs matched with city-level variables across 305 prefecture-level cities. By 2023, total hidden debt of city-level governments reached ¥119.388 billion, representing a hidden debt-to-GDP ratio of 42.66%. When combined with official on-balance-sheet debt, the total local government debt-to-GDP ratio rises to 66.93%.¹¹

Table 1 presents descriptive statistics for our city-year panel from statistical yearbooks. Hidden debt through LGFVs averages ¥109.84 billion per city, substantially exceeding average on-balance-sheet debt of ¥74.25 billion, highlighting the dominance of off-budget financing mechanisms. Large standard deviations in GDP (mean ¥438.77 billion) and bank loans (mean ¥750.47 billion) indicate substantial economic disparities across cities. Variables including GDP per capita, population, and housing prices also vary significantly, reflecting diverse local economic conditions.¹²

¹⁰Also known as urban investment bonds, or municipal corporate bonds, commonly referred to as Chengtouzhai in the Chinese literature.

¹¹These figures cover only city-level governments, excluding county and provincial debts. Actual total local government debt is likely substantially higher.

¹²Housing price data are sourced from 58.com, a major Chinese real estate platform. Land transaction prices from the *Yearbook of China's Land Resources* are excluded due to extensive missing observations.

Table 1: Descriptive Statistics for Main Variables (City-Year Panel, 2014–2023)

Variable	Mean	SD (Overall)	Between SD	Within SD	N (Obs./Cities)
On-Sheet LGD (Billion)	74.25	102.69	88.68	43.11	2,212 / 296
LGFVs Debt (Billion)	109.84	317.95	263.52	178.27	2,655 / 266
Blank Loan (Billion)	380.58	824.83	750.47	344.97	2,710 / 271
Budget Expenditure (Billion)	51.72	74.52	72.92	15.86	3,030 / 303
GDP (Billion)	308.02	452.78	438.77	114.32	3,030 / 303
GDP Per Capita (RMB)	61,549	35,358	32,122	14,879	3,030 / 303
Population (Million)	4.46	3.66	3.65	0.34	3,030 / 303
Housing Price (RMB/sqm)	8,424	6,623	6,057	1,904	2,427 / 288

Note: Summary statistics are based on an unbalanced panel of 303 cities from 2014 to 2023. Reported are the mean, overall, between, and within standard deviations. N refers to the number of non-missing city-year observations, with the number of cities in parentheses. Data sources: China City Statistical Yearbook, LGFVs debt data from Wind, and housing prices from 58.com.

Table 2 displays the ownership distribution of firms with outstanding corporate bonds from WIND. Local State-Owned Enterprises (LSOEs) constitute the majority at 86.63%, followed by Central State-Owned Enterprises (CSOEs) at 7.91% and Private Enterprises (POEs) at 5.47%. Among 4,182 total bond-issuing firms, 2,966 are classified as LGFVs¹³ (representing 82.7% of all state-owned bond issuers), while 320 are publicly listed companies (7.77% of total bond issuers).

Table 2: Distribution of Firms with Outstanding Bonds by Ownership Type

Firm Type	Overall (Freq, %)	Between (Freq, %)	Within (%)
Central State-owned Enterprises	2,652 (7.91%)	308 (7.51%)	100.00%
Local State-owned Enterprises	29,053 (86.63%)	3,586 (87.42%)	100.00%
Private Enterprises	1,833 (5.47%)	208 (5.07%)	100.00%
Total	33,538 (100.00%)	4,102 (100.00%)	100.00%

Note: Summary statistics are based on firm-level bond data from WIND. The sample includes 4,102 firms with outstanding bonds from 2014 to 2023. Among them, 2,966 are identified as LGFVs (82.7% of SOEs). A total of 325 listed firms (7.77% of all bond firms) are excluded from the final firm-level dataset.

Table 3 presents the ownership structure of publicly traded firms from CSMAR. Comparing ownership distributions between bond issuers and listed companies reveals substantial differences, particularly the underrepresentation of private enterprises in corporate bond markets.

¹³Of the 2,940 LGFVs successfully matched with city variables, 81.6% are state-owned enterprises.

This pattern suggests significant barriers to bond market access for private firms, consistent with state-directed credit allocation patterns in China’s financial system.

Table 3: Distribution of Listed Firms by Ownership Type

Firm Type	Overall (Freq, %)	Between (Freq, %)	Within (%)
Central State-owned Enterprises	3,390 (6.72%)	340 (6.74%)	100.00%
Local State-owned Enterprises	10,245 (20.31%)	1,025 (20.31%)	100.00%
Private Enterprises	36,805 (72.97%)	3,681 (72.95%)	100.00%
Total	50,440 (100.00%)	5,046 (100.00%)	100.00%

Note: Summary statistics are based on listed firm data from CSMAR. The sample includes 5,046 listed firms from 2014 to 2023. Among them, 5,007 are classified as non-financial firms, and approximately 4,000 are identified as industrial firms.

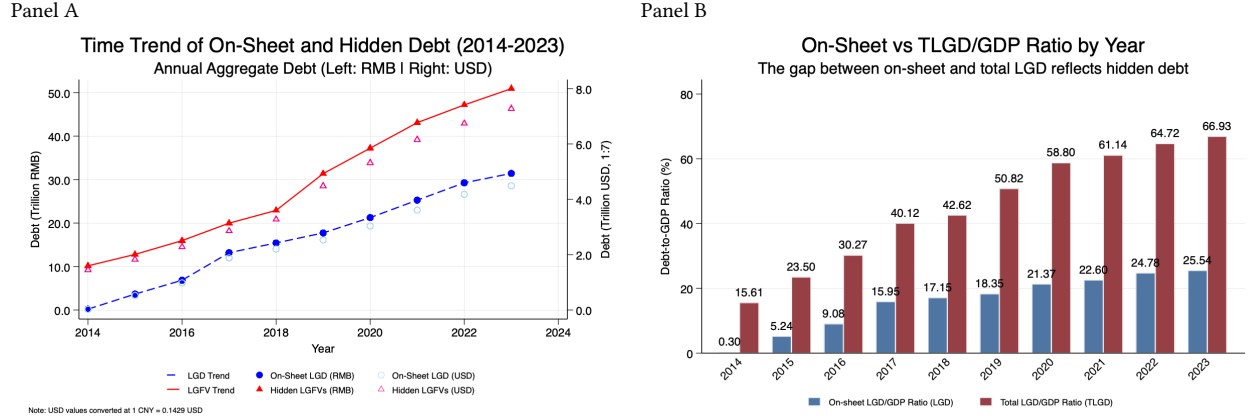
4 Validation

In order to validate the representativeness of our dataset, we aggregate the local government debt data to the national and provincial levels and compare the results with those from existing studies, demonstrating that our data is comparable. A detailed comparison of the provincial-level results with those of Ninghua Zhong is provided in the Appendix [A.4](#).

Figure [2](#) presents government debt trends from 2014 to 2023. Panel A shows steady growth in both on-sheet local government debt (LGD) and hidden debt through LGFVs, with hidden debt rising more sharply. Panel B reveals that the hidden debt-to-GDP ratio consistently surpasses the on-sheet ratio, driving the total debt-to-GDP ratio from 15.61% in 2015 to 66.93% in 2023. This widening gap highlights the growing macroeconomic significance of hidden liabilities and the need for greater transparency and oversight.

Figure [3](#) illustrates the composition and growth of China’s local government debt from 2014 to 2024, separated into on-sheet and off-sheet categories. On-sheet debt, shown in Panel A, is divided into *General Bonds* and *Special-Purpose Bonds (SPBs)*. General Bonds primarily finance public welfare projects, while SPBs are increasingly utilized for infrastructure projects and the

Figure 2: Overview of Government Debt Trends



Note: This figure presents the time trends of local government debt from 2014 to 2023, based on city-level data aggregated by year. Panel A shows the total stock of on-sheet and off-sheet (LGFV) debt. Panel B plots the corresponding debt-to-GDP ratios. Data sources: China City Statistical Yearbook and WIND.

refinancing of hidden debt. By 2024, the issuance of SPBs surged to RMB 31.42 trillion, reflecting their growing significance in local debt restructuring. Off-sheet debt, displayed in Panel B, consists of *Bonds Payable* and *Long-Term Loans* issued through Local Government Financing Vehicles (LGFVs). In 2024, Bonds Payable reached RMB 15.23 trillion, closely aligning with the National Audit Office's (NAO) hidden debt estimate of RMB 14 trillion.

5 The Rise of Local Government Debt

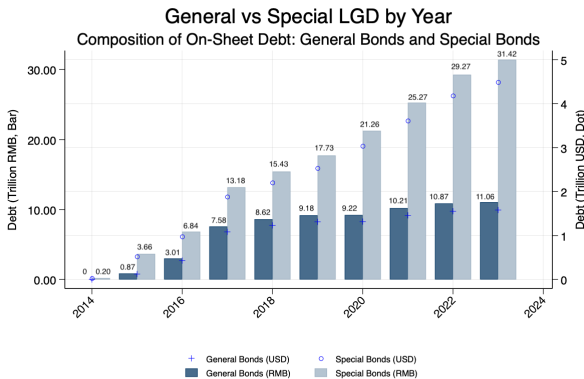
This section applies our city-level dataset to examine the geographical dynamics of local government debt in China. The city-level granularity enables precise mapping of debt accumulation patterns, revealing heterogeneity masked in provincial aggregates.

5.1 Rapid Growth and Geographic Patterns of Government Debt

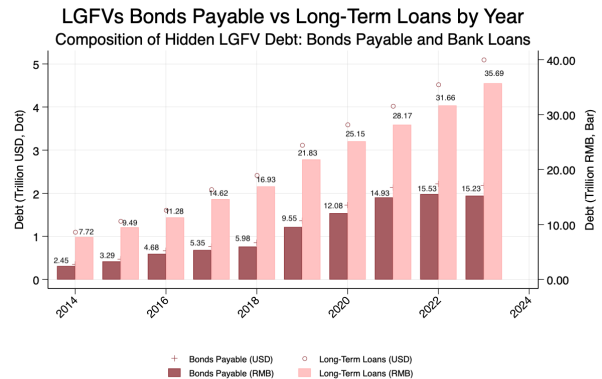
China's local government debt landscape transformed dramatically between 2015 and 2023, with off-balance-sheet financing emerging as the dominant driver of both growth magnitude and ge-

Figure 3: Overview of Government Debt Financial Composition Trends

Panel A



Panel B



Note: This figure presents the composition and evolution of China's local government debt from 2014 to 2023, based on city-level data aggregated by year. Panel A shows on-sheet debt, split into General Bonds and Special-Purpose Bonds (SPBs). Panel B displays off-sheet debt issued by Local Government Financing Vehicles (LGFVs), including Bonds Payable and Long-Term Loans. Data sources: China City Statistical Yearbook and WIND.

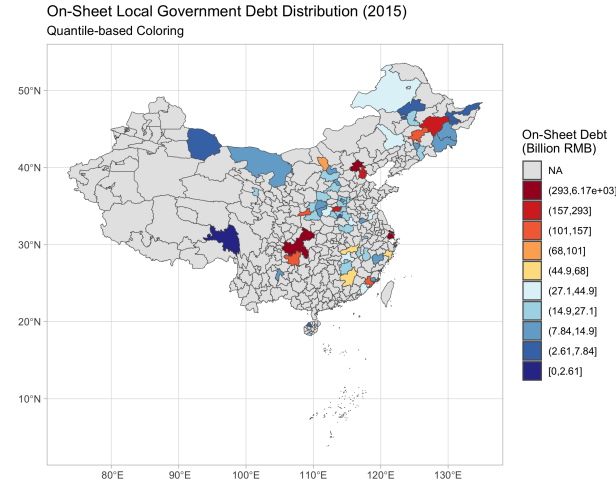
ographic dispersion.

Figure 4 demonstrates the scale of debt expansion across China's cities, with total debt balances increasing markedly in virtually all regions and the most dramatic changes occurring in eastern and central provinces. On-balance-sheet debt (Panels A-B) grew substantially in economically prominent cities like Beijing, Shanghai, and Guangzhou, reflecting these municipalities' enhanced capacity for formal bond issuance within official budgetary frameworks. However, the most striking transformation occurred in off-balance-sheet LGFV debt (Panels C-D), which expanded far more rapidly than formal debt and became particularly concentrated in mid-sized cities across provinces such as Henan, Anhui, and Hubei. The intensity of color changes between 2015 and 2023 in these panels reveals that off-balance-sheet liabilities not only accelerated faster but also became more geographically concentrated, creating distinct pockets of financial vulnerability.

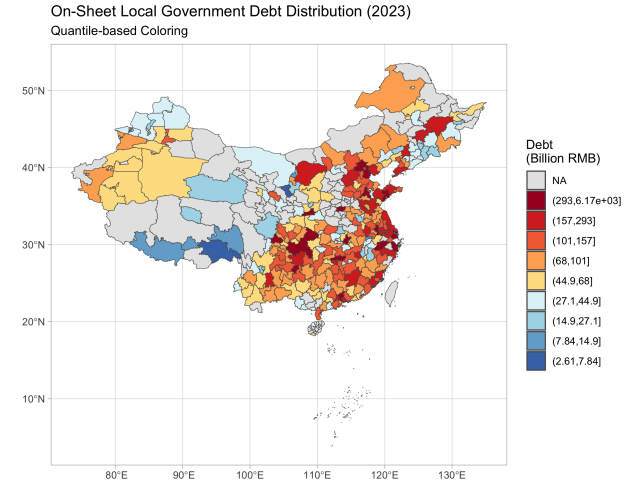
The debt-to-GDP ratios in Figure 5 reveal even starker geographic disparities and sustainability concerns. While on-balance-sheet debt ratios (Panels A-B) remained relatively stable across most cities, off-balance-sheet ratios (Panels C-D) surged dramatically, with the transfor-

Figure 4: Geographical Distribution of Debt Balances: 2015 vs. 2023

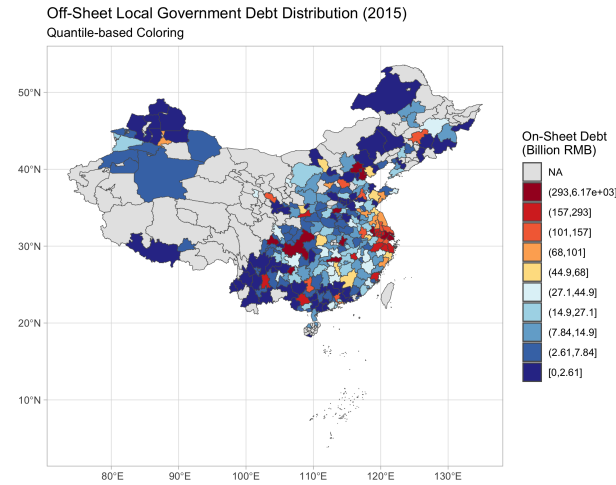
Panel A



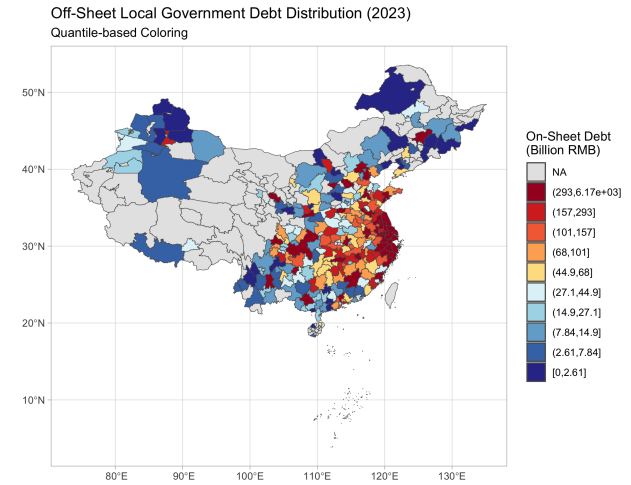
Panel B



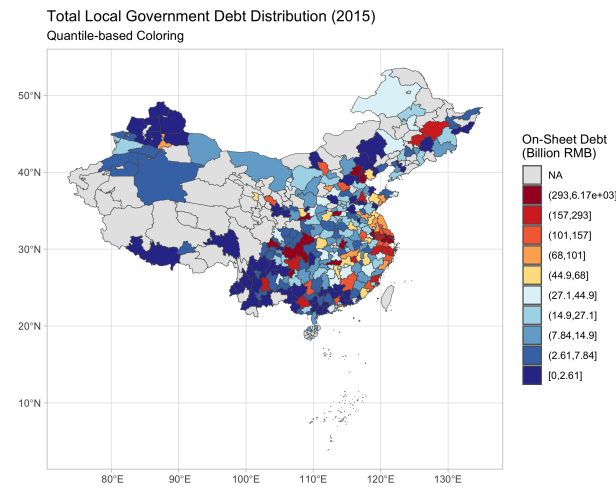
Panel C



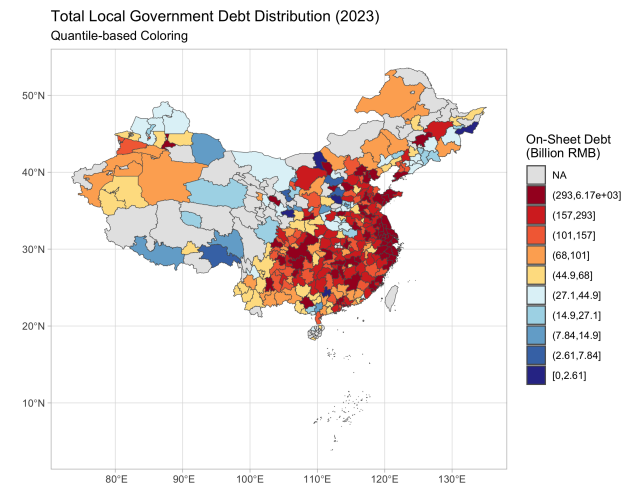
Panel D



Panel E



Panel F

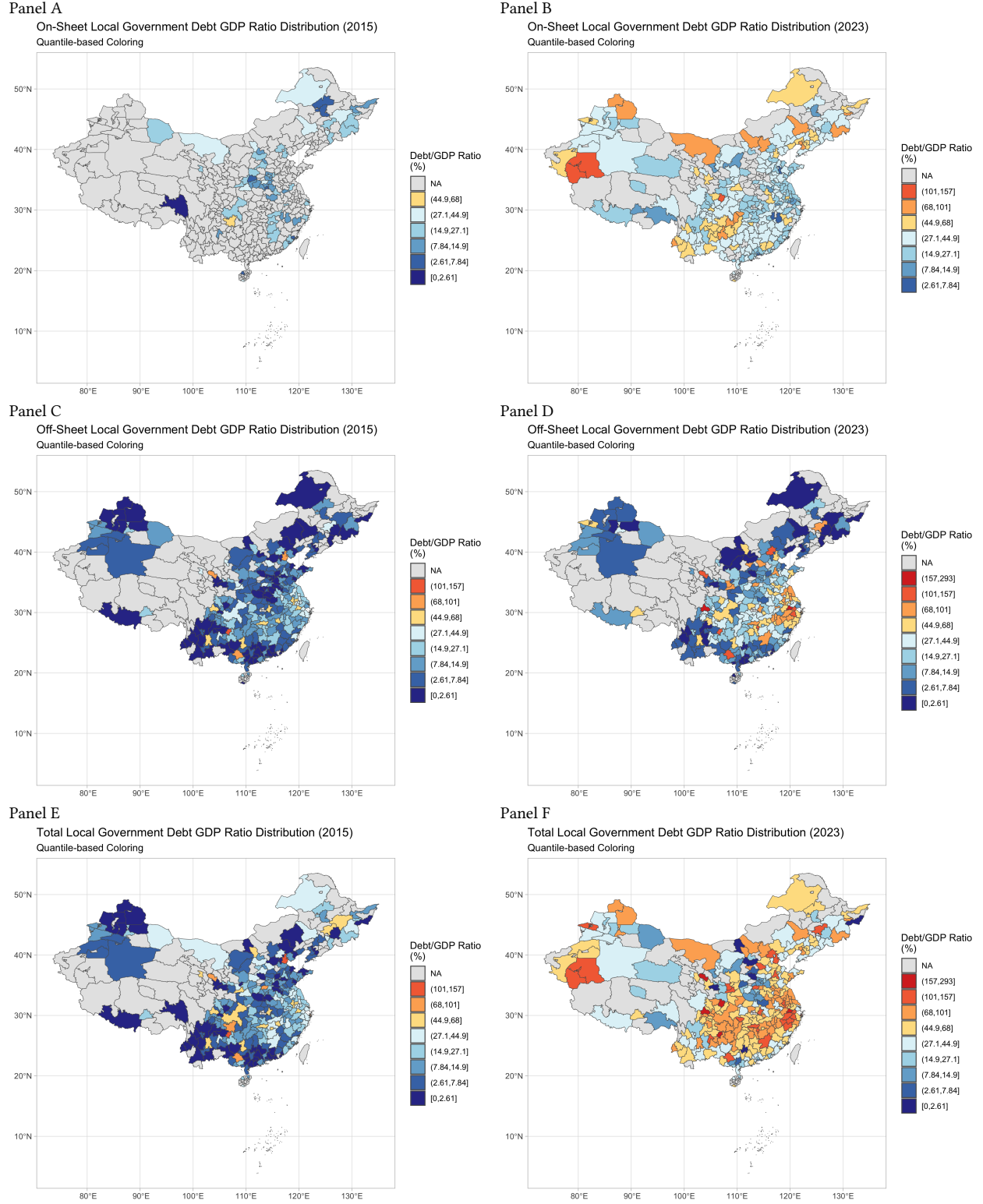


Note: This figure maps the geographical distribution of local government debt balance in 2015 and 2023 using city-level data. Cities are grouped into ten quantile bins based on the pooled distribution of 2015 and 2023 values, ensuring consistent color scales across years. Data sources: China City Statistical Yearbook and WIND.

mation particularly pronounced in central and southwestern regions where cities in provinces like Henan, Guizhou, and Anhui recorded sharp increases in hidden debt burdens. This divergence reflects underlying institutional constraints: coastal cities with stronger fiscal capacities maintained access to formal bond markets, while fiscally constrained inland regions increasingly relied on LGFV financing to sustain infrastructure investments. By 2023, total debt-to-GDP ratios (Panels E-F) highlight mounting solvency risks concentrated in mid-sized cities that lack both formal borrowing capacity and sufficient economic growth to service their obligations.

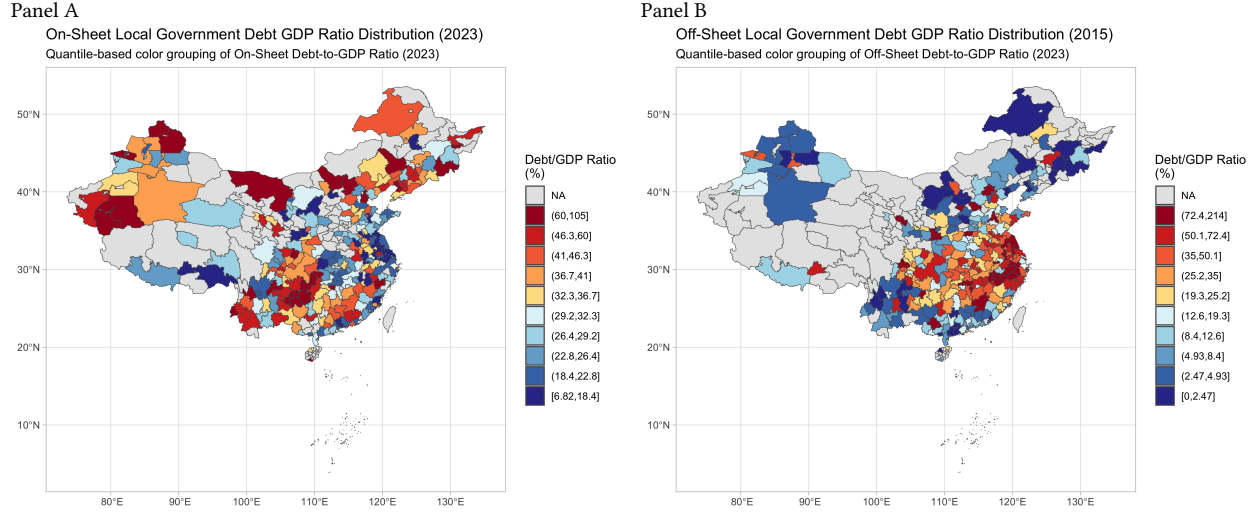
Figure 6 reveals the most important structural insight: a striking inverse relationship between formal and informal debt channels. Cities with high on-balance-sheet debt ratios consistently exhibit low off-balance-sheet debt levels, and vice versa, demonstrating systematic substitutability between financing mechanisms rather than complementary use. The geographic distribution shows that regions facing institutional barriers to formal borrowing systematically turned to LGFV channels, while fiscally stronger cities prioritized regulated bond markets. This substitution pattern explains why off-balance-sheet debt became the primary driver of both the magnitude and geographic variation of local government debt expansion, creating distinct dual-high debt zones that have become focal points of China's debt restructuring initiatives.

Figure 5: Geographical Distribution of Debt/GDP Ratio: 2015 vs. 2023



Note: This figure maps the geographical distribution of local government debt/GDP ratio in 2015 and 2023 using city-level data. Cities are grouped into ten quantile bins based on the pooled distribution of 2015 and 2023 values, ensuring consistent color scales across years. Data sources: China City Statistical Yearbook and WIND.

Figure 6: Cross-City Comparison of Debt Channels Using Quantile-Based Coloring (2023)



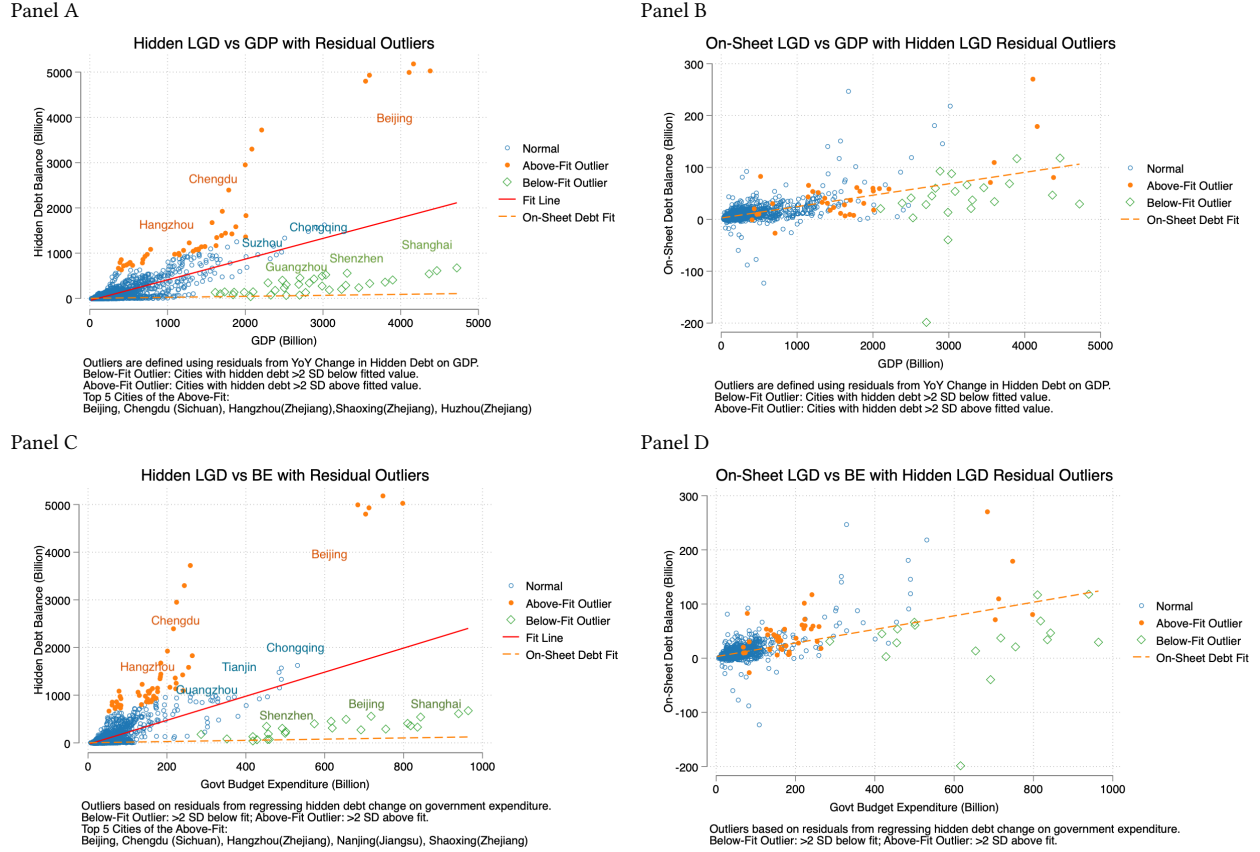
This figure compares on-sheet and off-sheet local government debt-to-GDP ratios across cities in 2023. Panel A maps on-sheet debt (LGD), and Panel B maps off-sheet LGFV debt (LGFVD). Cities are grouped into ten quantiles based on each variable's own cross-sectional distribution, enabling relative ranking and spatial comparison. The inverse pattern across panels suggests a substitutive relationship between formal and off-balance-sheet financing. Data sources: China City Statistical Yearbook and WIND.

5.2 Debt and Economic Fundamentals

Having established the geographic patterns of debt expansion, we now examine how debt levels correlate with economic fundamentals to understand the drivers of debt accumulation and substitution between financing channels.

Figure 7 reveals stark differences between hidden and on-balance-sheet debt relationships and provides direct evidence of substitution effects. Hidden debt shows strong positive correlation with city GDP (Panel A), with cities identified as above-fit outliers (more than 2 standard deviations above the fitted line) including Beijing, Chengdu, Hangzhou, and other major municipalities carrying disproportionately high hidden debt. Crucially, when examining these same cities in the on-balance-sheet debt relationship (Panel B), they distribute both above and below the fitting line rather than clustering as high outliers, demonstrating that high hidden debt does not translate to uniformly high formal debt. This pattern provides direct visual evidence of the

Figure 7: Scatter Plots of Hidden and On-Sheet Debt vs City Economic Indicators

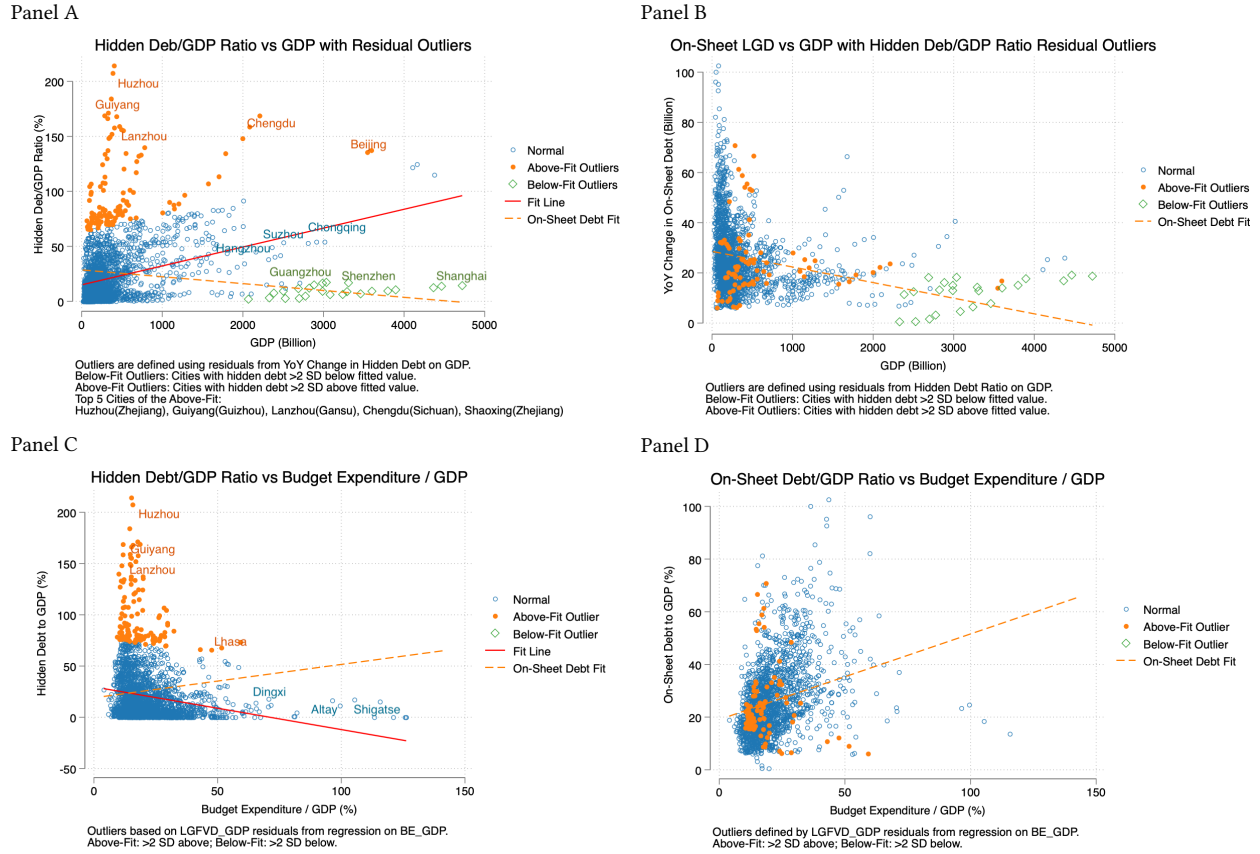


Note: This figure presents scatter plots relating city-level hidden and on-sheet debt balances to economic indicators, based on the city-level dataset constructed in this paper. A clear substitution pattern emerges across all panels: cities with unusually high hidden debt often exhibit relatively lower on-sheet debt levels. Data sources: *China City Statistical Yearbook* and WIND.

substitution mechanism. Shanghai and Guangzhou exemplify below-fit outliers with relatively lower hidden debt despite their large economies. The budget expenditure relationships (Panels C-D) reinforce this substitution pattern, with cities like Tianjin and Chongqing showing similar divergent positions across the two debt channels.

Figure 8 provides normalized measures that illuminate sustainability patterns and fiscal constraints. Hidden debt-to-GDP ratios are systematically higher in smaller cities (Panel A), with cities like Huzhou, Guiyang, Lanzhou, and Shaoxing identified as above-fit outliers exhibiting exceptionally high debt burdens relative to their economic size. Larger economies generally maintain lower debt-to-GDP ratios, suggesting better debt sustainability. The substitution effect

Figure 8: Scatter Plots of Debt-to-GDP Ratios vs City-Level Economic Fundamentals



This figure presents scatter plots relating city-level debt-to-GDP ratios to economic fundamentals, based on the city-level dataset constructed in this paper. Panels A and B show that hidden debt ratios tend to be higher in smaller cities, while on-sheet debt ratios remain more stable across the income distribution. Panels C and D examine debt ratios relative to budget expenditure, revealing a substitution pattern: cities with constrained budgets often rely more on hidden debt. Overall, the plots illustrate variation in debt composition and fiscal strategies across Chinese cities. Data sources: *China City Statistical Yearbook* and WIND.

becomes more pronounced in the ratio analysis: when examining on-balance-sheet debt ratios (Panel B), the cities with high hidden debt ratios show more dispersed positioning rather than consistently high formal debt ratios. The budget expenditure analysis (Panels C-D) exposes the underlying fiscal dynamics, with cities facing budget constraints (lower budget expenditure relative to GDP) relying more heavily on hidden financing, as evidenced by the negative slope in Panel C, while on-balance-sheet debt correlates positively with budget capacity (Panel D).

These patterns establish four critical findings. First, hidden debt substantially dominates the

debt landscape across most cities. Second, economic scale provides protection through lower debt-to-GDP ratios. Third, systematic substitution effects create an inverse relationship where hidden debt compensates for limited formal borrowing access. Fourth, fiscal constraints drive off-balance-sheet reliance, creating a spiral where cities least equipped to service debt become most dependent on hidden financing. These relationships reveal that China’s debt challenge stems from systematic migration to less transparent financing channels driven by institutional and fiscal constraints. Additional scatter plots with economic indicators are provided in Appendix A.5.

6 A Micro Anatomy of LGFVs and Hidden Debt

This section presents an application of the newly constructed firm-level dataset, focusing on Local Government Financing Vehicles (LGFVs). Utilizing detailed financial and operational information, we document several distinguishing features of LGFVs, including their financial structures, industry distribution, and the evolution of key financial indicators over time. This analysis highlights just one of the many potential uses of the comprehensive dataset developed in this study.

Table 4: Comparison of Firm-Level Financial Variables: Non-LGFV SOEs vs. LGFVs

Variables	Non-LGFV SOEs		LGFVs		Mean diff SOEs - LGFVs
	Obs/N.	Mean	Obs/N.	Mean	
Leverage (%)	12,991 / 1,557	53.23	23,700 / 2,947	53.23	0.00
Investment Rate (%)	11,896 / 1,458	2.71	19,718 / 2,774	3.64	-0.93
Return on Assets (%)	12,939 / 1,554	2.09	23,677 / 2,945	0.92	1.17***
Total Assets (Billion)	13,048 / 1,558	57.41	23,732 / 2,947	42.39	15.02***
Total Liabilities (Billion)	12,992 / 1,557	42.66	23,700 / 2,947	25.05	17.61***
Total Revenue (Billion)	10,543 / 1,218	11.03	2,855 / 338	0.02	11.01***
Net Profit (Billion)	12,955 / 1,556	0.96	23,709 / 2,946	0.25	0.71***
Long-Term Loans (Billion)	10,063 / 1,425	6.42	23,230 / 2,929	8.73	-2.31***
Short-Term Loans (Billion)	10,910 / 1,451	2.70	19,494 / 2,878	1.32	1.38***
Bonds Payable (Billion)	6,334 / 1,289	8.90	18,803 / 2,900	4.74	4.17***

Note: This table summarizes key financial characteristics of Local Government Financing Vehicles (LGFVs) and non-LGFV state-owned enterprises (SOEs), based on firm-level panel data constructed from Wind and CSMAR. All monetary values are in billion RMB. “Obs/N.” refers to the number of firm-year observations and the number of unique firms, respectively. Asterisks indicate statistical significance of mean differences based on two-sample t-tests: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

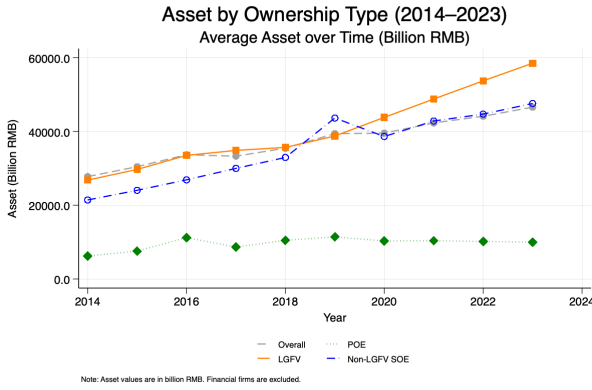
Table 4 compares key firm-level financial variables between Local Government Financing Vehicles (LGFVs) and non-LGFV state-owned enterprises (SOEs), highlighting several important distinctions. LGFVs exhibit slightly higher leverage ratios and significantly larger total assets and liabilities, reflecting their larger scale of operations despite tighter margins. Although their leverage is similar to that of non-LGFV SOEs, LGFVs benefit from lower effective interest rates, likely due to their quasi-governmental status and implicit support from local governments, which enables them to secure more favorable financing terms. Moreover, LGFVs tend to generate lower returns on assets, suggesting that their primary objective is not profit maximization but rather the fulfillment of policy mandates such as infrastructure development and public service projects. This is further evidenced by their heavier reliance on bond financing, consistent with their role in raising capital for local governments rather than operating purely as profit-driven enterprises.

Figure 9 presents firm-level financial trends from 2014 to 2023, highlighting significant differences between Local Government Financing Vehicles (LGFVs), non-LGFV state-owned enterprises (SOEs), and privately-owned enterprises (POEs). Throughout this period, LGFVs exhibited notably higher average total assets and liabilities compared to other firms, reflecting their larger operational scale and substantial financial obligations. Regarding leverage (total liabilities to assets ratio), LGFVs consistently maintained elevated levels, surpassing both non-LGFV SOEs and POEs, emphasizing their heavy reliance on debt financing. Conversely, investment activities of LGFVs showed notable volatility but generally remained high, likely associated with their role in infrastructure projects and public services. Despite their scale and investment intensity, LGFVs displayed persistently lower weighted average profitability and return on assets (ROA), indicative of their policy-driven rather than profit-oriented mandates. This analysis represents just one illustrative application of the comprehensive firm-level dataset; additional time trend analyses by other firm characteristics can be found in the Appendix A.6.

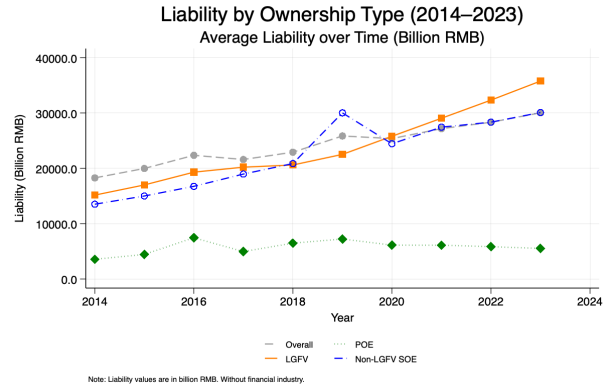
Figure 10 presents the city-level geographic distribution of unique Local Government Financing Vehicles (LGFVs), constructed using the newly developed firm-level dataset. The map illustrates significant heterogeneity in LGFV presence across Chinese cities. Particularly, regions

Figure 9: Local Government Financial Vehicles

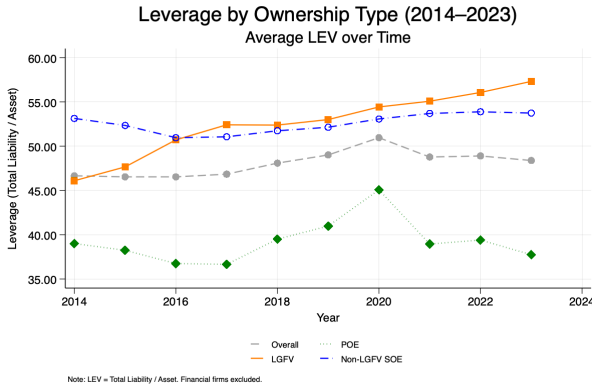
Panel A



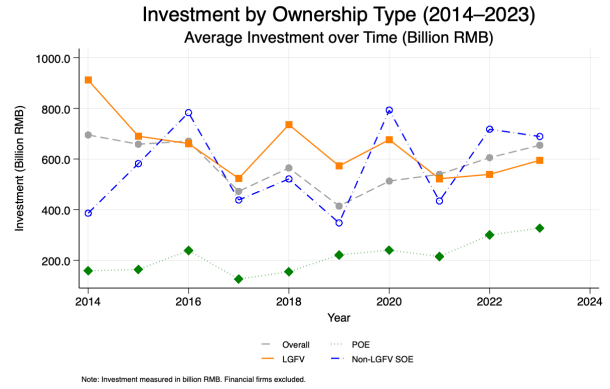
Panel B



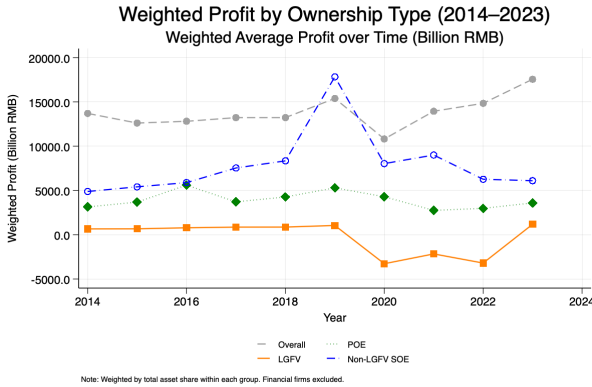
Panel C



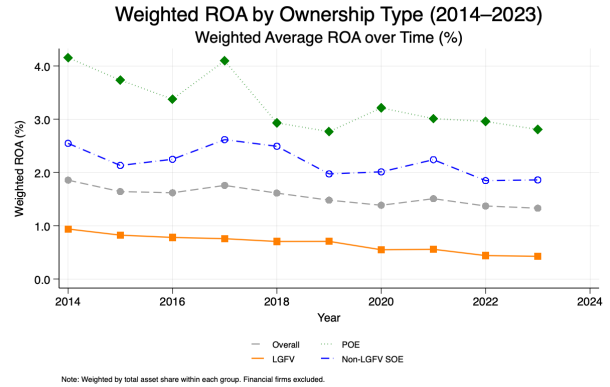
Panel D



Panel E



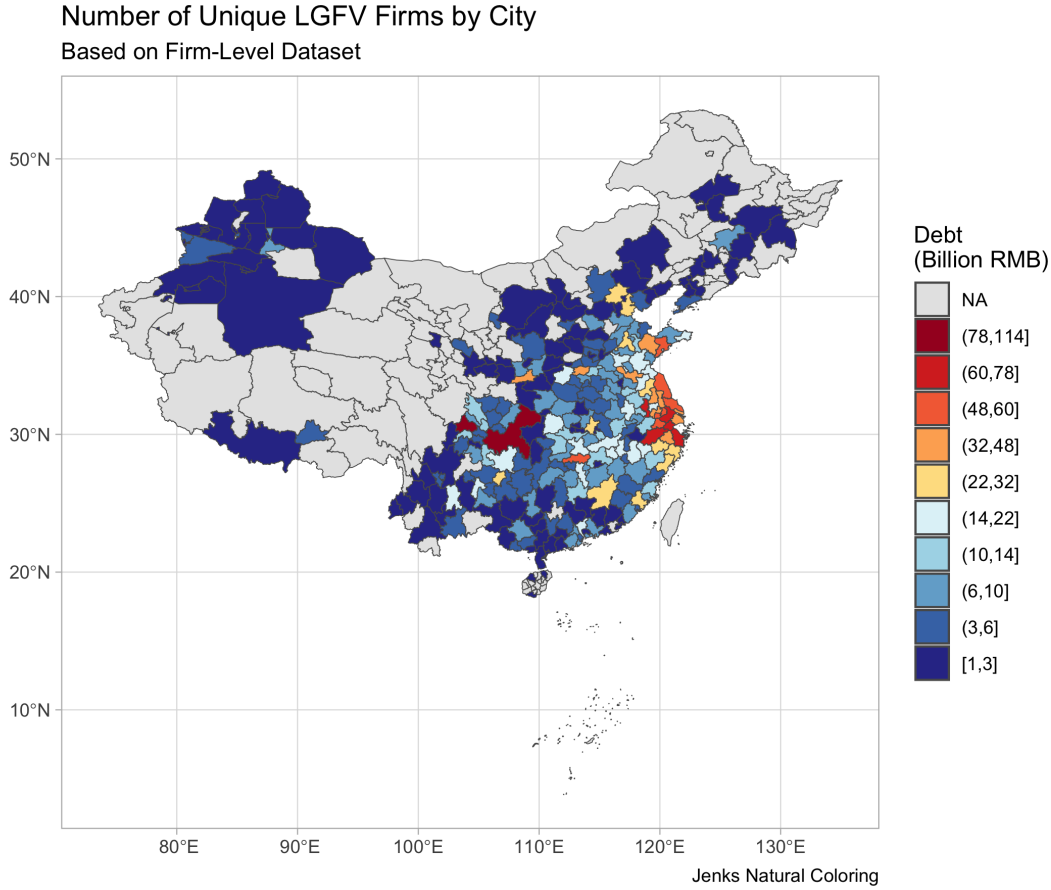
Panel F



Note: This figure presents the time trends of firm-level financial indicators for LGFVs, non-LGFV SOEs, and POEs from 2014 to 2023 using firm-level panel data. Weighted values are weighted by total assets within each group. Data sources: Wind and CSMAR. Additional time trend analyses by other firm characteristics can be found in the Appendix A.6

along the eastern coastal areas and selected inland economic hubs exhibit a notably higher concentration of LGFVs. For instance, cities such as Chengdu (city code: 5101) and Chongqing (city

Figure 10: Spatial Distribution of Local Government Financing Vehicles (LGFVs)



Note: This figure maps the spatial distribution of unique LGFVs across Chinese cities as of August 2024, based on our firm-level dataset. City-level counts reflect the number of distinct LGFVs identified by location. Cities are grouped using Jenks natural breaks classification. Data sources: Wind

code: 5000) have the highest counts of unique LGFVs, suggesting their substantial engagement in off-sheet debt financing and infrastructure development. Conversely, other regions show considerably fewer LGFVs, reflecting differential local fiscal capacities or varied infrastructure investment needs. This map represents a preliminary application of the firm-level dataset developed in this paper. The dataset supports a wide range of city- and firm-level investigations into LGFVs and holds substantial potential for integration with other data sources. Further LGFV-related analyses based on this dataset are presented in Appendix [A.7](#).

7 Conclusion

This paper constructs and introduces a novel firm-level dataset encompassing 2,940 industrial Local Government Financing Vehicles (LGFVs) across 305 Chinese cities from 2014 to 2023. By systematically integrating detailed financial metrics, city-level fiscal data, and macroeconomic indicators, this dataset provides unprecedented granularity to investigate China's hidden local government debt.

The descriptive analyses highlight several key stylized facts. First, hidden off-sheet debt through LGFVs has expanded dramatically, significantly outpacing officially recorded on-sheet debt. By 2023, the hidden debt reached approximately RMB 50.9 trillion, pushing the total local debt-to-GDP ratio to nearly 67%. Second, notable regional disparities in debt accumulation patterns emerged, with inland and economically less-developed cities disproportionately reliant on hidden borrowing. Third, firm-level analysis indicates LGFVs operate with consistently higher leverage and lower profitability, underscoring their primary role as vehicles for implementing local government policies rather than pursuing profit maximization.

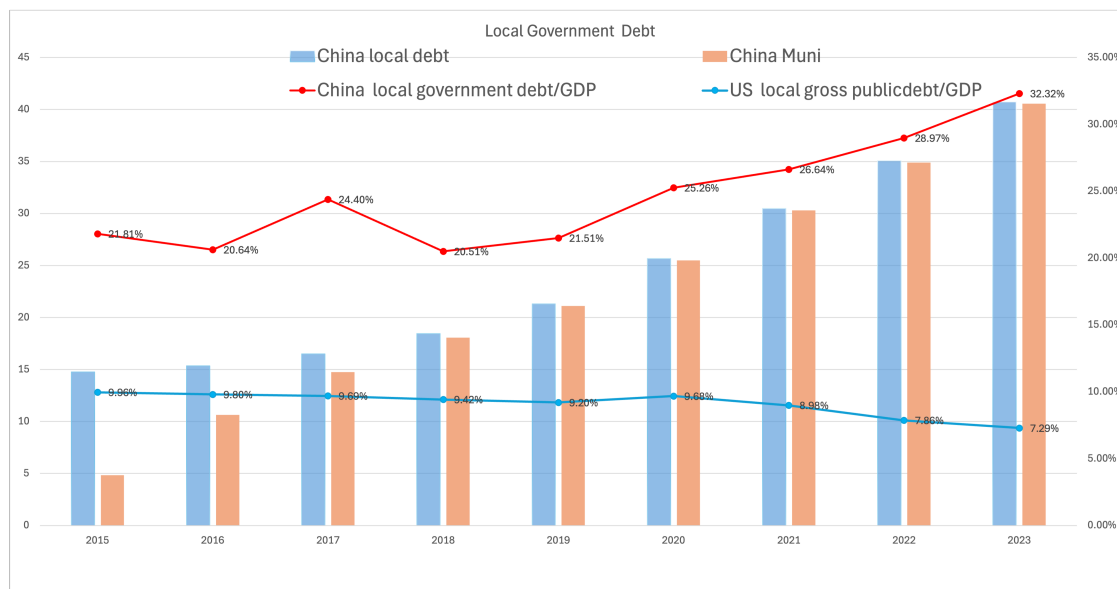
The dataset created by this study fills a critical data gap and lays the empirical groundwork for future research into the determinants, impacts, and systemic risks associated with hidden local government debt. Further studies might explore the interactions between fiscal transparency, local political incentives, real estate dynamics, and credit allocation efficiency. Ultimately, by enhancing transparency and facilitating evidence-based policymaking, this work contributes to a deeper understanding of fiscal sustainability challenges in China's rapidly evolving economic landscape.

bib

A Appendix

A.1 Comparison of U.S. and Chinese on-sheet municipal bonds

Figure 11: On-Sheet Local Government Debt: China vs. U.S. (2015–2023)



Note: This figure compares U.S. and Chinese on-balance-sheet local government debt trajectories. Data source: China's debt data are from the [China Electronic Local Government Bond Market Access platform](#); U.S. data from the [Budget of the United States Government](#).

A.2 Firm-level and City-level variables

Table 5 presents the Firm-level variables we have, with "Issuer Rating at Issue" being the only variable specific to bond firms, the firms with outstanding bonds (from WIND). Table 6 presents the City-level variables we have.

A.3 China Local Government Debt

Chinese local governments have relied on multiple borrowing channels to finance infrastructure and development since the 2008–09 stimulus. A key challenge in estimating local government

Table 5: Firm-Level Data Description

Variable Description	Variable Code
Firm ID	firm_ID
Industry Name	CSRC
Industry Code	CSRC_maj
City	city_code
Province	pro_code
Enterprise Type	type_firm
Inception Date	InceptionDate
Listed Status	Listed
Issuer Rating at Issue	rate_firm
Total Assets	asset
Net Fixed Assets	FA
Fixed Assets Disposal	disposalFA
Total Invested Capital	capital
Interest-Bearing Liabilities	IBL
Total Liabilities	liability
Bonds Payable	BP
Long-Term Loans	LTL
Short-Term Loans	STL
Net Profit	profit
Total Revenue	income
Net Profit Attributable to Parent Company	Eprofit
Comprehensive Income Attributable to Parent Company	Eequity
Interest Expenses	fincost
Taxes and Surcharges	tax
Depreciation of Fixed Assets	depre
Monetary Funds	cash

debt is measuring *off-sheet local government debt* (also termed off-balance-sheet debt, hidden debt, or LGFV debt), which represents the primary focus of this analysis. Tables 7 and 8 report the yearly total (sum) of on-sheet and off-sheet local government debt, respectively, based on my constructed dataset. Figure 12 provides an overview of the evolution of hidden debt and LGFVs based on the dataset constructed in this paper.

Table 6: City-Level Variable Description

Variable Description	Variable Code
City Code (4 digits)	city_code
Year	year
Province Code (2 digits)	pro_code
Province Name	pro
City Name	city
GDP (Billion RMB)	GDP
Secondary Industry Output (Billion RMB)	GDP_2
On-Balance-Sheet Local Government Debt (Billion RMB)	LGD
General On-Balance-Sheet LG Debt (Billion RMB)	LGDg
Specialized On-Balance-Sheet LG Debt (Billion RMB)	LGDs
GDP Per Capita (RMB)	GDP_PC
GDP Index (Base Year = 100)	GDPindex
Housing Price (RMB/sqm)	LP
Population (Million)	POP
Bank Loan Balance (Billion RMB)	BL
Local Government Budget Revenue (Billion RMB)	BR
Local Government Budget Expenditure (Billion RMB)	BE
Hidden LGFV Debt (Billion RMB)	LGFVD
Total Local Government Debt (Billion RMB)	TLGD
Leverage Ratio (Debt / Asset) for Firm Group d	LEV_d
City-Level Aggregate of Variable X for Firm Group d	X_d

Note: Firm group d refers to categories such as LGFVs, LSOEs, POEs, and their listed/unlisted status.

Off-sheet local government debt refers to liabilities not officially recorded in local governments' budgetary statements but for which local governments bear ultimate repayment responsibility through legal or implicit guarantees. These obligations exceed statutory borrowing limits and are often backed by fiscal funds or informal government guarantees (?). During the post-2009 credit boom, local governments increasingly relied on off-balance-sheet borrowing via LGFVs, trusts, and other shadow channels to circumvent direct borrowing restrictions (?). While off-sheet debt theoretically includes various shadow financing channels, implicit government debt through non-LGFV channels is extremely difficult to measure. Therefore, this paper operationally defines off-sheet local government debt as debt issued through LGFVs, specifically bonds payable and long-term borrowing in LGFVs' liabilities. Since LGFVs are not required to disclose finan-

Table 7: On-Sheet Local Government Debt Summary

Year	Total LGD (Trillion)	General LGD (Trillion)	Special LGD (Trillion)	LGD to GDP Ratio (%)
2014	0.196	0.000	0.000	0.307
2015	3.608	0.858	1.041	5.339
2016	6.707	2.947	1.968	9.189
2017	12.637	7.155	5.058	15.764
2018	14.819	8.127	6.641	16.980
2019	17.012	8.641	8.358	18.132
2020	20.331	8.732	10.875	21.046
2021	24.207	9.651	14.270	22.295
2022	28.074	10.211	18.007	24.492
2023	30.278	10.431	20.031	25.361

Note: This table reports the annual on-sheet local government debt (LGD) amounts, separated into general and special LGD categories, from 2014 to 2023. The last column shows the ratio of total LGD to national GDP.

Table 8: Off-Sheet Local Government Debt Summary

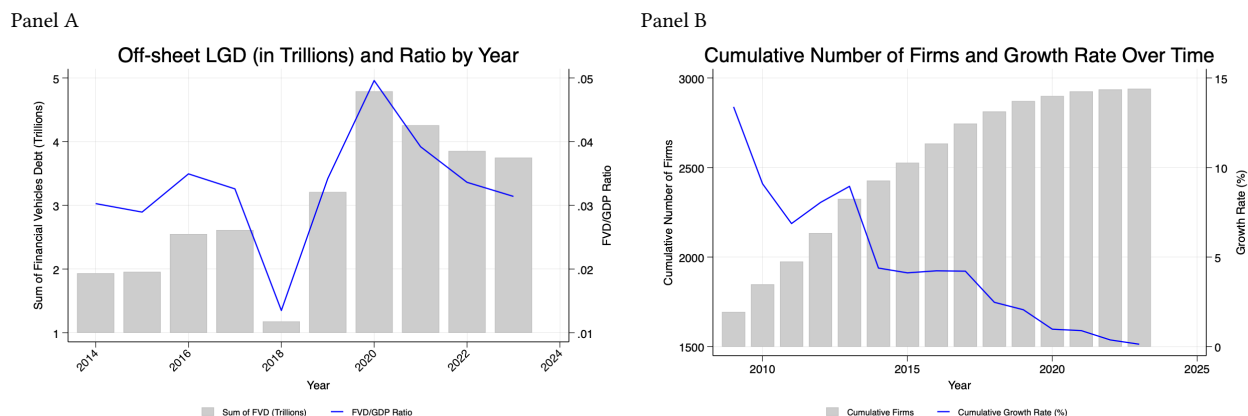
Year	Bonds Payable	Long-Term Loan	Total Debt	GDP	Debt to GDP Ratio (%)
2014	2.445	7.716	10.161	63.896	15.903
2015	3.290	9.492	12.781	67.577	18.914
2016	4.677	11.284	15.961	72.989	21.868
2017	5.355	14.623	19.978	80.163	24.922
2018	5.981	16.934	22.916	87.272	26.258
2019	9.547	21.829	31.376	93.826	33.440
2020	12.076	25.154	37.229	96.599	38.540
2021	14.929	28.169	43.098	108.579	39.693
2022	15.528	31.657	47.185	114.624	41.165
2023	15.234	35.693	50.928	119.388	42.657

Note: This table reports estimated off-sheet local government debt (LGFV debt) from 2014 to 2023, including bonds payable and long-term loans. All monetary values are in billion RMB. The debt-to-GDP ratio is calculated using total off-sheet debt over national GDP.

cial information publicly, research on local government debt relies on bond issuance data (?). This study defines **LGFVs as state-owned enterprises that issue municipal bonds for local government investment purposes**. Our identification process follows three steps: (1) identify municipal bonds using classification criteria, (2) extract LGFV lists from bond issuers, (3) collect

financial information for identified entities. We focus on outstanding industrial bonds, with financial data covering the bond issuance year plus three previous years (per disclosure regulations requiring four-year financial history).

Figure 12: Overview of Hidden Debt and LGFVs Trends



Note: This figure presents trends in estimated hidden local government debt and the number of LGFVs. Panel A displays the total local government debt (on-sheet plus off-sheet) as a share of GDP, while Panel B shows the annual number of unique LGFVs. Data sources: Wind and China City Statistical Yearbook.

A.3.1 Alternative Definitions of Municipal Bonds

A key challenge in studying LGFV debt involves definitional ambiguity regarding the terms "LGFV" and "municipal bond." Two main approaches prevail in both policy discussions and academic research: the audit-based definition from the National Audit Office (NAO) and the market-based definition exemplified by the ChinaBond classification.

NAO (Audit-Based Definition) The National Audit Office (NAO) approach relies on public market information and official government documents to identify LGFV debt. This conservative methodology, utilized by studies such as [huang2020local] in combination with CICC¹⁴ data, recognizes LGFV obligations only when supported by explicit documentation. The approach

¹⁴China International Capital Corporation (CICC) is China's first joint-venture investment bank, founded in 1995, offering comprehensive financial services including investment banking, fixed income, and asset management.

tends to capture primarily newer LGFVs established after 2018, as post-2014 regulatory reforms prohibited local governments from providing explicit guarantees for LGFV debts. Consequently, this audit-based definition typically underreports the true scope of hidden local government debt and offers limited analytical granularity due to infrequent and aggregate national auditing practices.

ChinaBond (Market-Based Definition) The ChinaBond classification system, developed by the ChinaBond Pricing Center ¹⁵, adopts a more comprehensive market-based approach. This methodology, employed by studies such as [chen2020financing], incorporates information from both public and nonpublic channels to identify LGFV bonds based on their functional role in municipal financing. Available since 2013, the ChinaBond definition captures implicit government backing expectations and includes a significantly broader universe of LGFVs compared to official NAO estimates. This approach provides detailed financial information and frequent data updates essential for granular empirical analysis, reflecting market participants' understanding of implicit governmental support rather than relying solely on explicit legal guarantees.

This study adopts the ChinaBond market-based definition due to its practical advantages in data availability and analytical robustness. Employing this broader classification ensures comprehensive measurement of hidden local government liabilities, consistent with an expanding body of economic research. Thus, throughout this paper, "LGFVs" and "municipal bonds" refer specifically to entities and instruments identified via the ChinaBond classification.

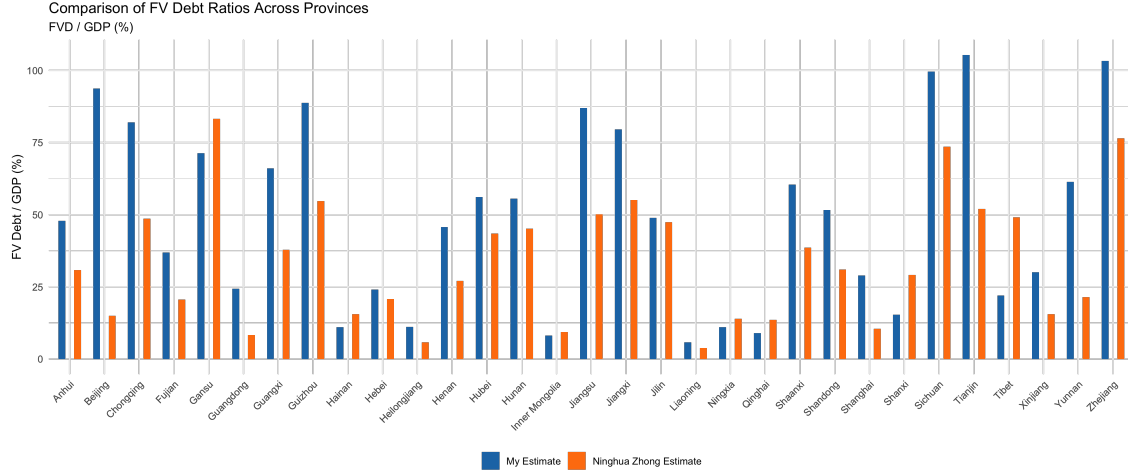
A.4 Hidden Debt Representative

To validate the robustness and credibility of our dataset, we compare our definition and measurement of financial vehicle debt (FVD) with those employed in Ninghua Zhong (2022). Figures 13

¹⁵ChinaBond Pricing Center, established in 2017, is a wholly owned subsidiary of China Central Depository & Clearing Co., Ltd. (CCDC). It provides authoritative pricing, yield curves, valuations, and bond indices used widely in financial markets.

to 15 present a side-by-side comparison of our estimates across provinces. As shown in Fig-

Figure 13: Financial Vehicles' Debt to GDP Ratio



ure 13, both our FVD-to-GDP ratios and Zhong's estimates exhibit similar provincial patterns, but also reveal important discrepancies in levels. These differences stem from several methodological distinctions. First, our bond data are sourced from the Wind database, whereas Zhong relies on the iFinD platform. Second, we define FVD more broadly as the sum of bond payables (BP) and interest-bearing liabilities (LBL), while Zhong's definition appears to focus more narrowly on outstanding bond amounts. Third, although our dataset captures current debt levels, it lacks complete historical issuance data, a limitation not present in Zhong's dataset. Fourth, our sample is restricted to industrial LGFVs, potentially excluding more diversified entities included in Zhong's estimation. Lastly, I scale debt by sector-level GDP (industrial or secondary), while Zhong uses total GDP, which may lead to additional variation in debt-to-GDP ratios. Figure 14 extends this comparison to total local government debt (TLGD), combining both on-sheet and off-sheet components. Again, our estimates are broadly comparable with Zhong's, though variations in coverage and definitions remain.

Figure 15 illustrates the provincial geographical distribution of estimated LGFV debt ratios in 2022. The left panel shows our TLGD measure (on-sheet + off-sheet), while the right panel visualizes Zhong's estimate based solely on LGFV bond data. Both maps capture similar regional

Figure 14: Total Local Government Debt to GDP Ratio

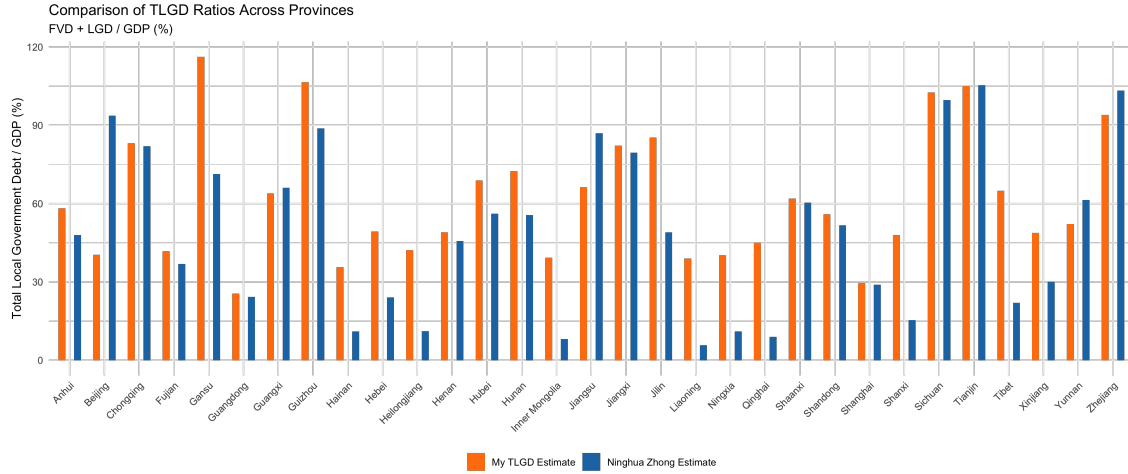
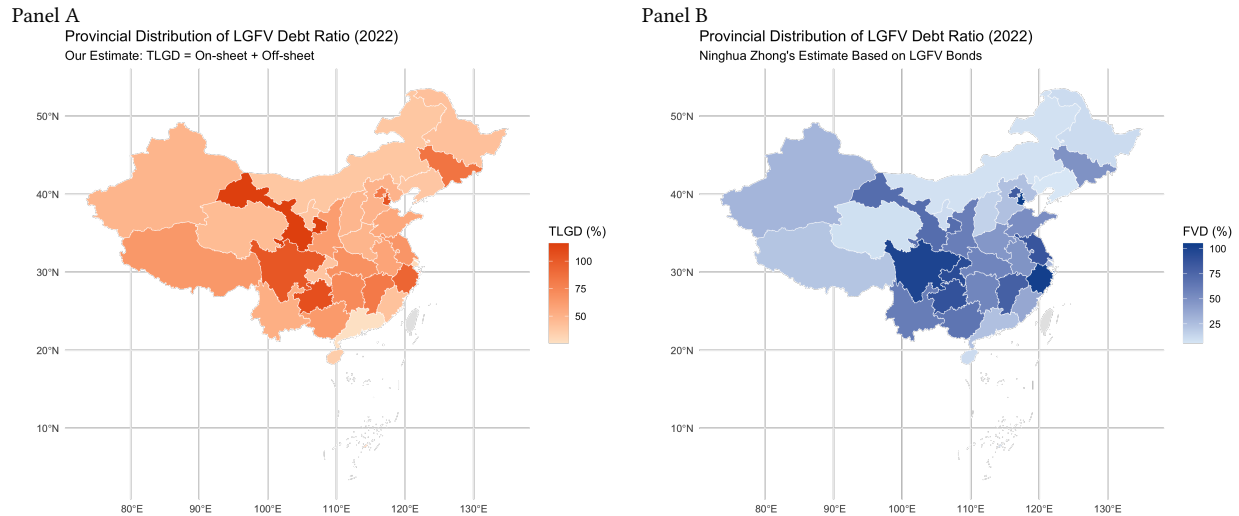


Figure 15: Geographical Distribution: This Paper vs. Ninghua Zhong (2022)



Note: This figure compares the spatial distribution of total local government debt (Panel A, based on this paper's estimates) and off-sheet LGFV debt from Zhong (2022) (Panel B). The maps highlight regional concentration patterns in local debt levels. Data sources: Wind, China City Statistical Yearbook, and Zhong (2022).

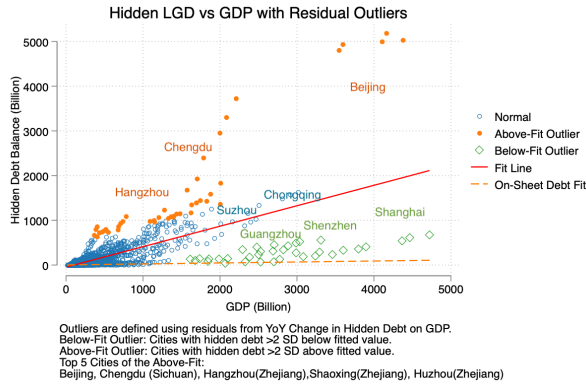
trends—particularly higher hidden debt burdens in central and southwestern provinces—but our results suggest a more extensive presence of off-sheet debt across regions.

Together, these figures demonstrate that despite definitional and data-source differences, our estimates of LGFV debt ratios are broadly consistent with those in existing studies, supporting the reliability of our constructed dataset.

A.5 City-Level Correlates: Debt and Economic Fundamentals

A.5.1 Debt Balance v.s. Indicator Balance

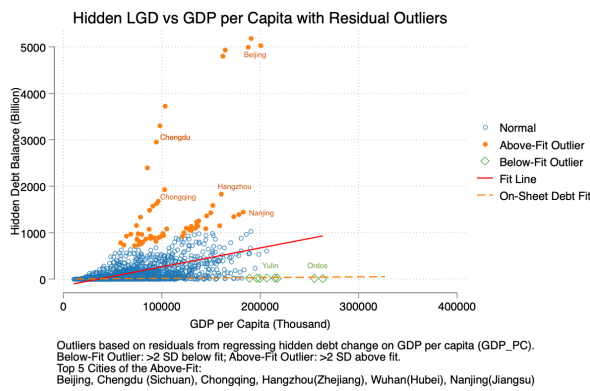
Panel A



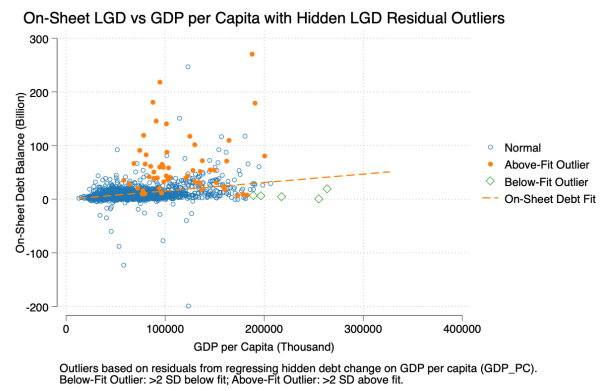
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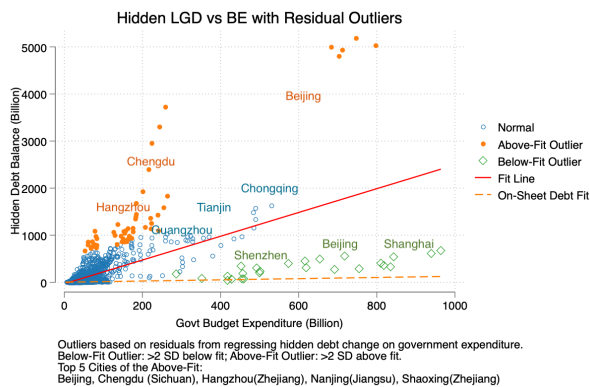
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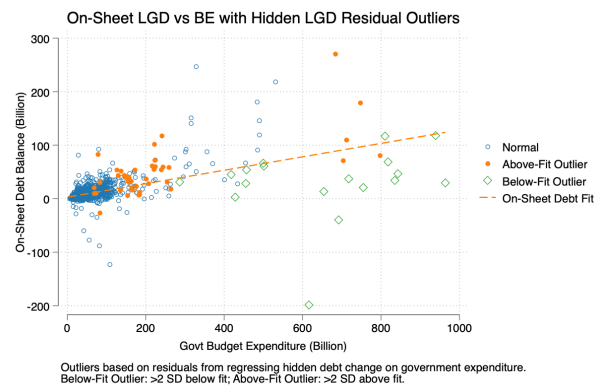
Panel D



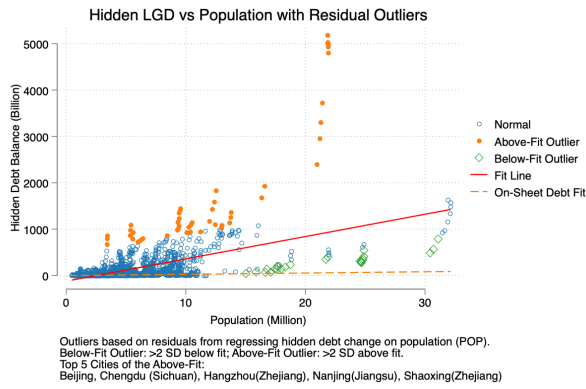
Panel E



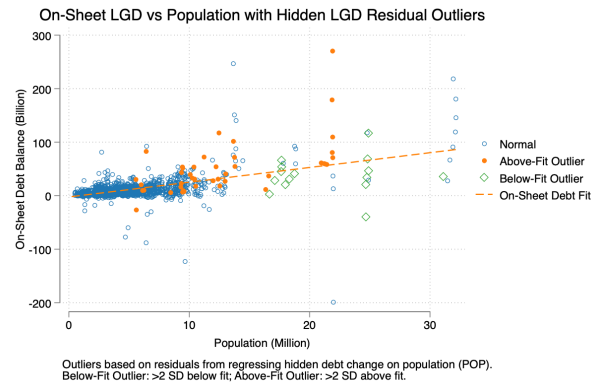
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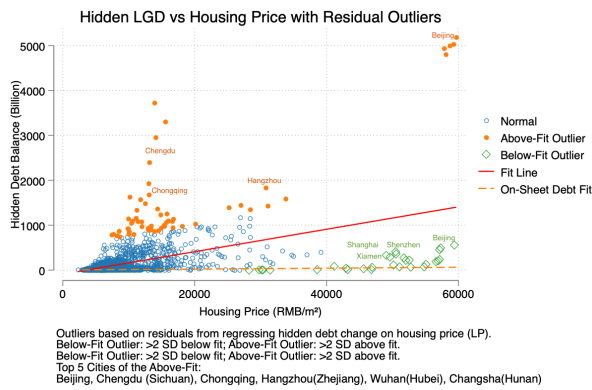
Panel G



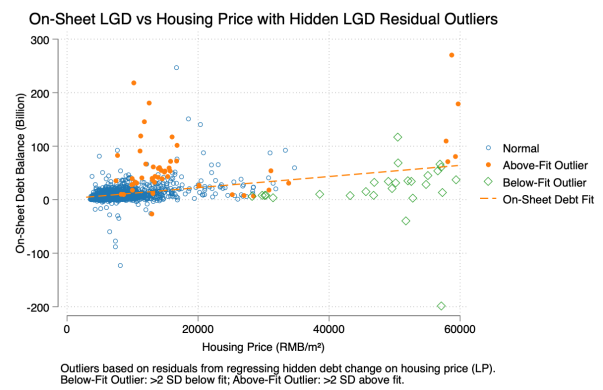
Panel H



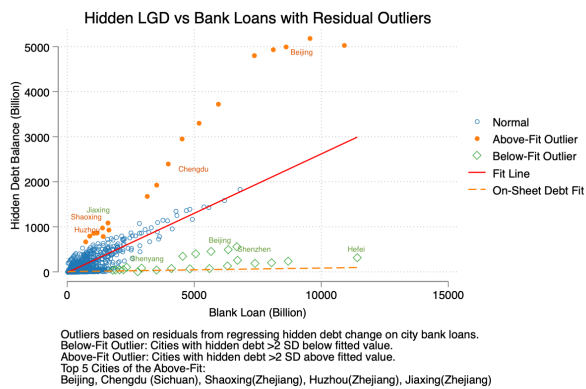
Panel I



Panel J



Panel K



Panel L

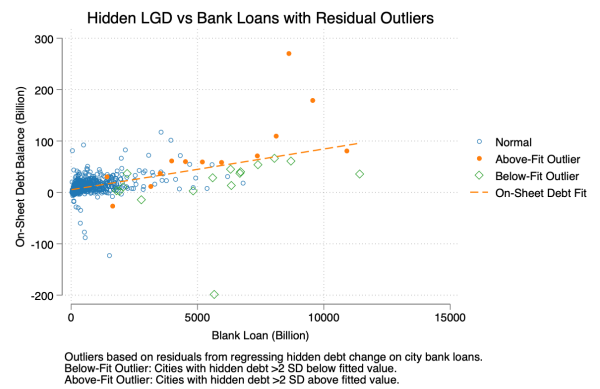
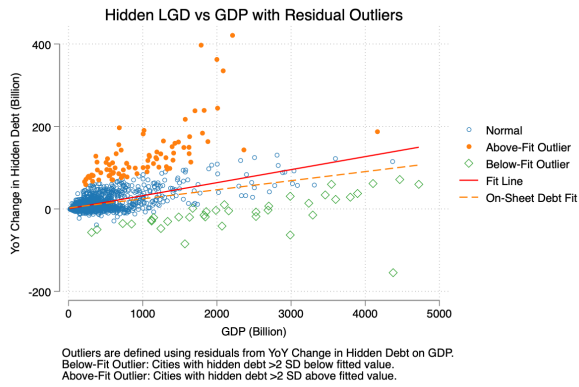


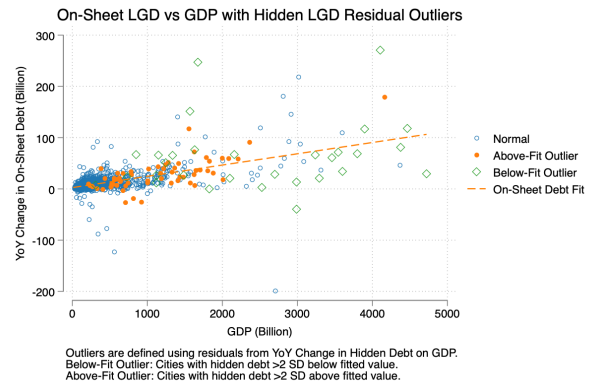
Figure 16: Hidden Debt (Balance) and On-Sheet Debt (Blance) across Various City Indicators.

A.5.2 Debt Change v.s. Indicator Balance

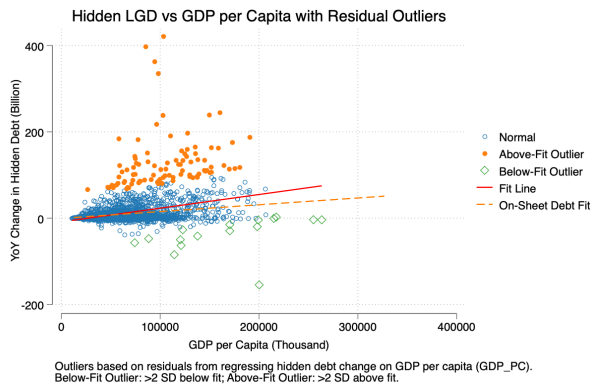
Panel A



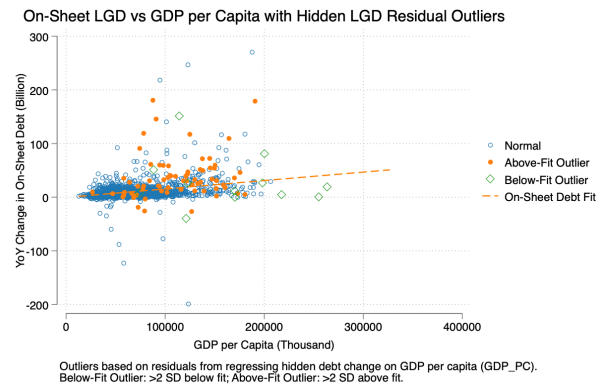
Panel B



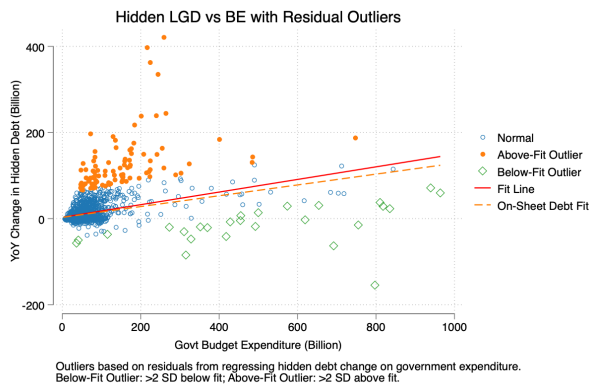
Panel C



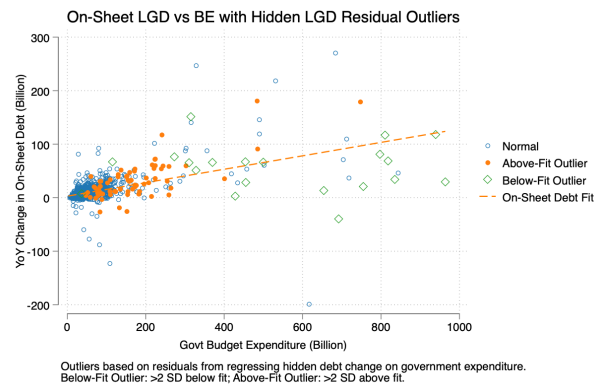
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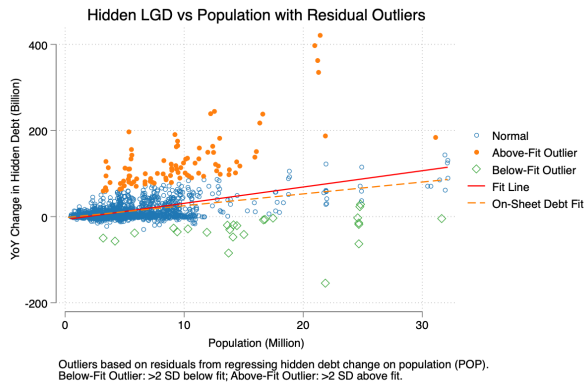
Panel E



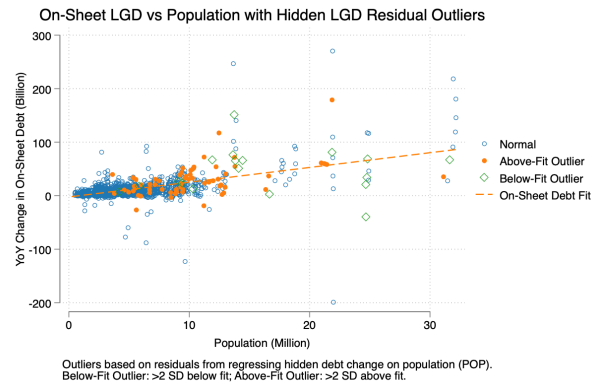
Panel F



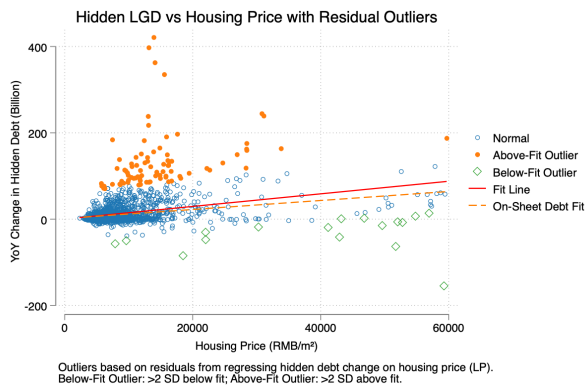
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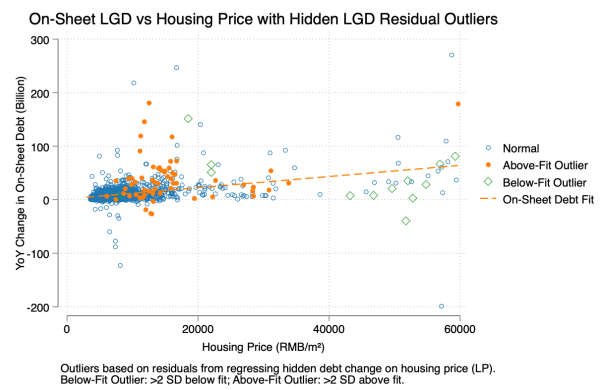
Panel H



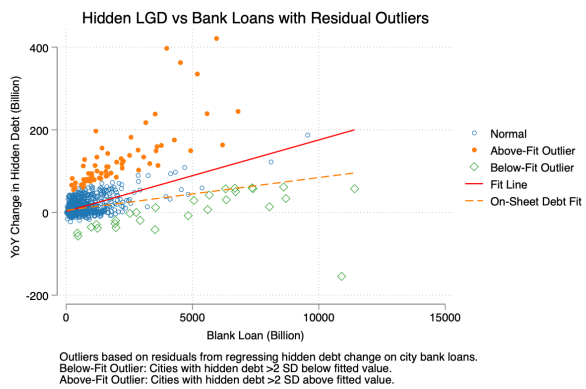
Panel I



Panel J



Panel K



Panel H

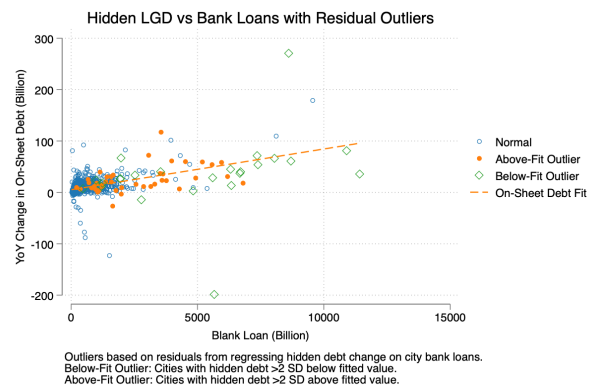
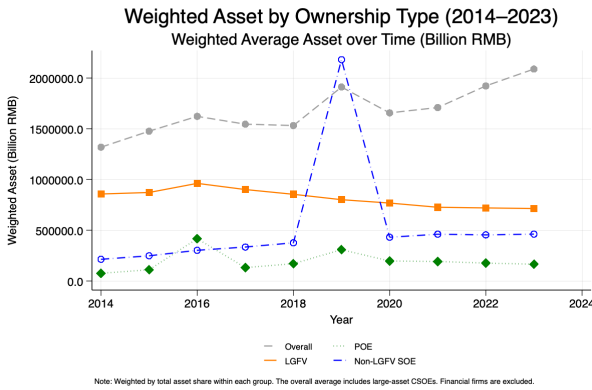


Figure 17: Changes in Hidden Debt and On-Sheet Debt against City-Level Indicators.

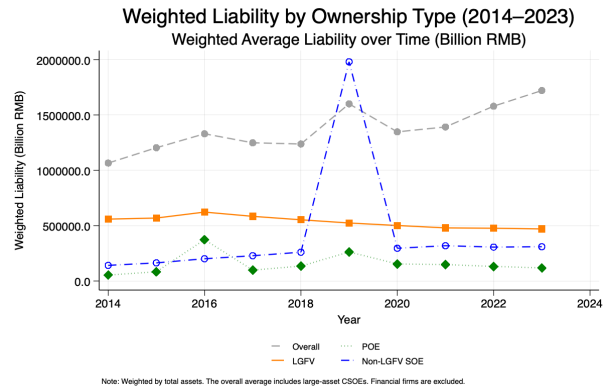
A.6 Additional Financial Performance time trend

Figure 18: Local Government Financial Vehicles Performance Time Trend

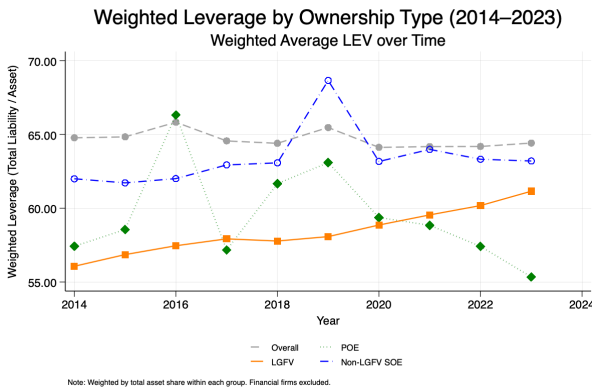
Panel A



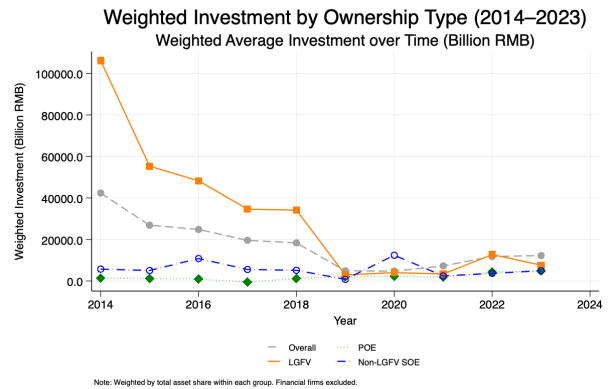
Panel B



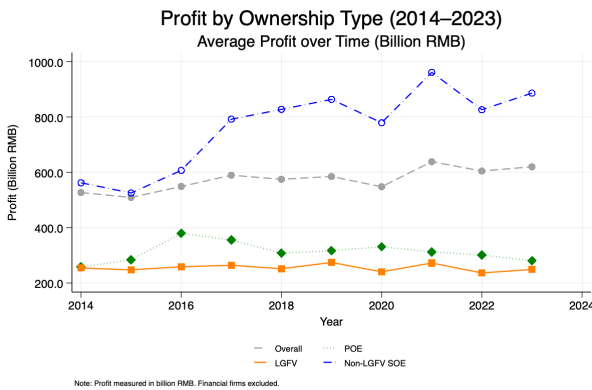
Panel C



Panel D



Panel E



Panel F

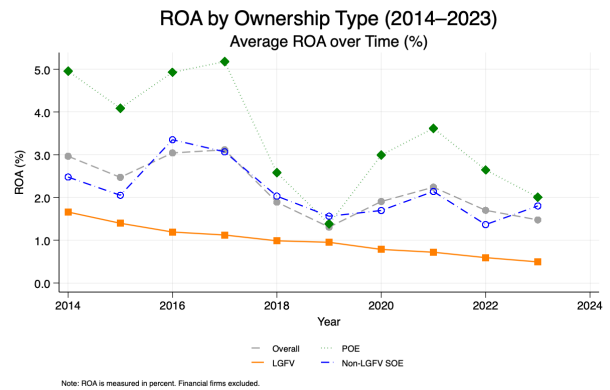
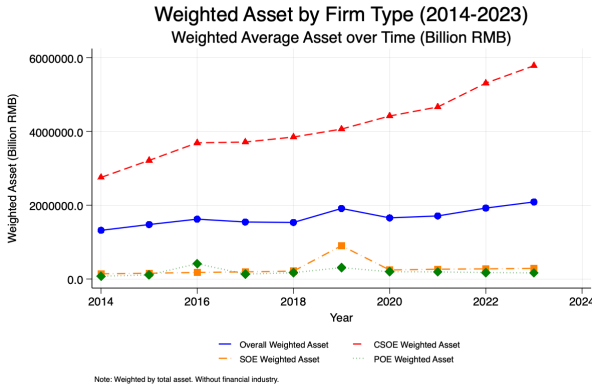
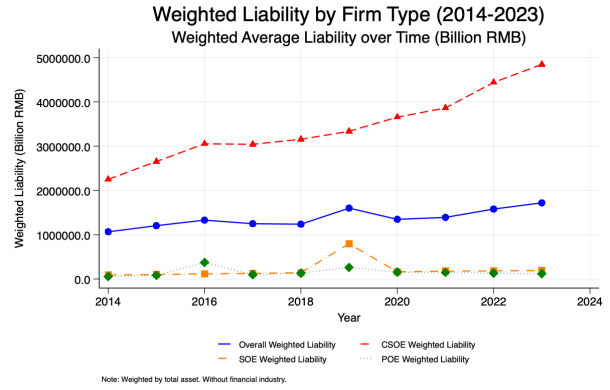


Figure 19: Firms Financial Performance Time Trend

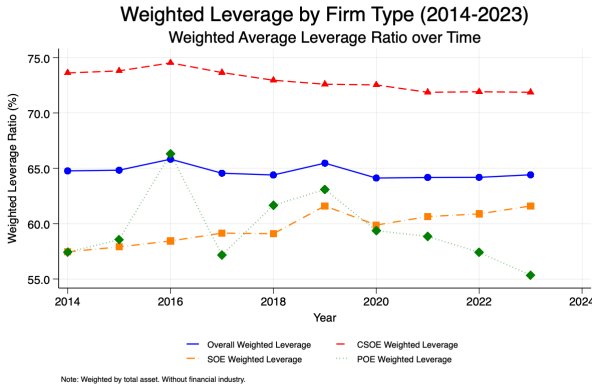
Panel A



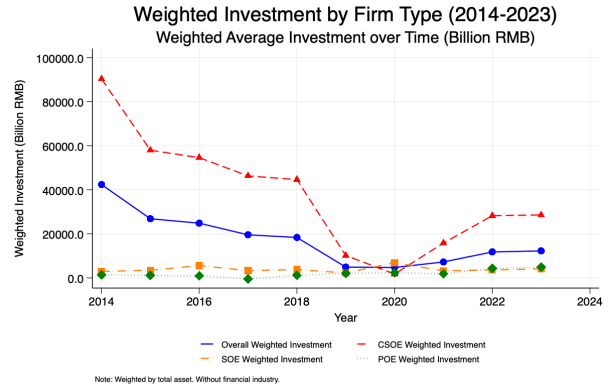
Panel B



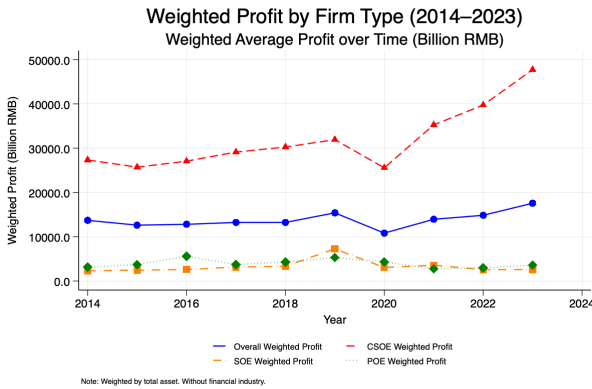
Panel C



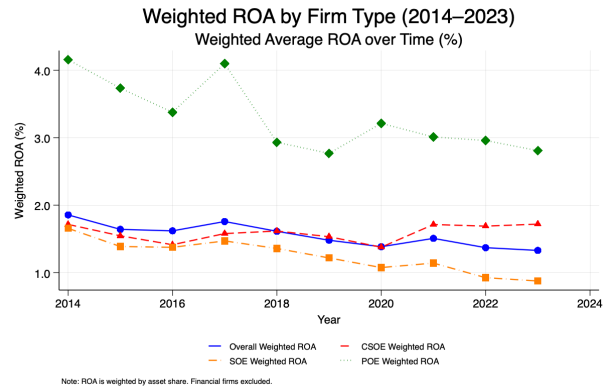
Panel D



Panel E



Panel F

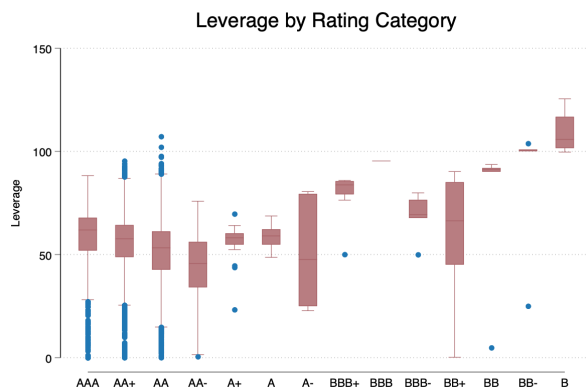


A.7 Additional Analysis on LGFVS

Figure 20 presents the rating distribution based on leverage for both LGFVs and non-LGFV SOEs, while Table 9 summarizes the return on assets (ROA) across different rating categories for LGFVs. In Panel A of Figure 20, LGFVs exhibit high leverage levels alongside strong issuer ratings, high-

lighting their dominant position in the bond market. In contrast, Panel B shows that non-LGFV SOEs are more sensitive to leverage when determining their ratings. Moreover, Table 9 reveals that ROA for LGFVs does not vary markedly across rating categories, suggesting that factors such as market positioning and the implicit support from local governments play a more pivotal role in shaping their credit profiles than profitability.

Panel A. Local Government Financial Vehicles (LGFVs)



Panel B. Non-LGFV State-Owned Enterprises (SOEs)

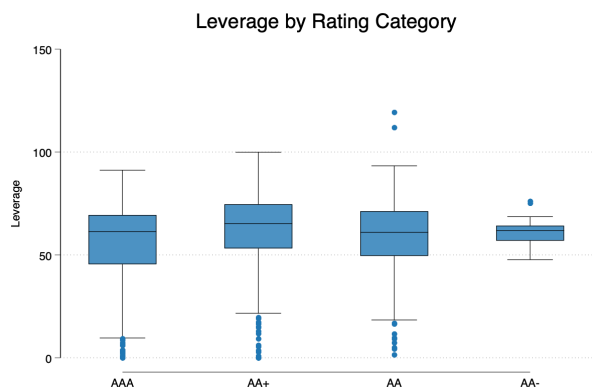


Figure 20: Rating Distribution on Leverage in 2023: LGFVs and Non-LGFV SOEs

Table 9: Descriptive Statistics of ROA (%) by Credit Rating

Rating	Mean	SD	Min	Max	N
AAA	0.723	1.039	-5.151	14.760	2357
AA+	0.742	0.916	-13.099	20.814	6789
AA	1.030	0.970	-20.983	19.110	12648
AA-	1.372	1.081	-2.326	16.402	1132
A+	0.923	0.445	0.440	1.810	17
A	0.424	0.513	0.089	1.765	10
A-	-0.086	0.252	-0.486	0.270	7
BBB+	-0.120	0.210	-0.465	0.079	9
BBB	-0.092	0.064	-0.137	-0.047	2
BBB-	0.541	0.718	-0.261	1.426	4
BB+	-0.329	0.665	-2.505	0.132	22
BB	-1.127	0.488	-1.498	-0.271	5
BB-	14.055	34.106	-3.872	75.000	5
B	-8.029	14.409	-29.410	0.854	4
Total	0.929	1.117	-29.410	75.000	23011

Figure 21 shows the industry distribution of LGFV versus non-LGFV SOE bond issuers in

2023. LGFVs concentrate heavily in infrastructure sectors, with construction dominating at 1,819 issuers compared to only 113 non-LGFV SOEs. In contrast, non-LGFV SOEs predominate in commercial sectors like manufacturing (501 vs. 2 LGFVs). This pattern reflects LGFVs' primary function as financing vehicles for local public infrastructure, distinguishing them from commercially-oriented SOEs.

Figure 21: Industry Distribution of LGFV vs Non-LGFV SOE Bond Issuers, 2023

